

The Compass



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Community Mathematics Centre

Schools of the Krishnamurti Foundation India
(KFI)

Statement of intent

The magazine is published by the Community Mathematics Centre. It is a collaborative initiative of the schools of the Krishnamurti Foundation India (KFI).

Its intention is:

- To provide material for mathematics and science teachers who teach at the high school or middle school levels, and for students who study at these levels.
- To serve as a platform where teachers and students can share their findings and discoveries with others who share their love for the subject.

In addition to mathematical content, in each issue we shall share a few ‘snippets’ from the world of science — short extracts abstracted from articles in online magazines, selected on the basis of intrinsic interest.

The Compass is published as an online magazine free of cost. Articles may be downloaded and used freely by teachers, but with proper attribution.

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Colour coding

The magazine has articles addressed to teachers who teach at different levels: Lower Primary level; Upper Primary and High School levels; and Senior Secondary and Junior College levels. There is also a Problem Corner. In addition we have the Snippets.

For the convenience of the reader, we use the following colour coding to distinguish between the various levels. Of course, such a division can never be exact. So, our apologies in advance for errors of judgement!

- Primary & Upper Primary (Classes 4, 5, 6, 7)
- Upper Primary & High School (Classes 6, 7, 8, 9, 10)
- Senior Secondary & Junior College (Classes 10, 11, 12)
- Problem Corner
- Snippets

Fast Calculations – 4

We continue our study of computational shortcuts (or, “the art of doing fast calculations and impressing an audience”). We shall also explain the logic behind the shortcuts given as problems in the October 2024 issue.

Multiplying two numbers close to and on the same side of the same power of 10

It is easier to explain the procedure through examples than to describe it in general!

Example: To compute the product 97×96

- Both 97 and 96 are less than and close to 100, with $97 = 100 - 3$ and $96 = 100 - 4$.
- Cross-subtract: $97 - 4 = 93 = 96 - 3$.
- Multiply the two deficits: $3 \times 4 = 12$.
- Hence the answer is 9312.

Example: To compute the product 998×993

- Both 998 and 993 are less than and close to 1000, with $998 = 1000 - 2$ and $993 = 1000 - 7$.
- Cross-subtract: $998 - 7 = 991 = 993 - 2$.
- Multiply the two deficits: $2 \times 7 = 14$.
- Hence the answer is 991014.

Example: To compute the product 105×103

- Both 105 and 103 are greater than and close to 100, with $105 = 100 + 5$ and $103 = 100 + 3$.
- Cross-add: $105 + 3 = 108 = 103 + 5$.
- Multiply the two increments: $5 \times 3 = 15$.
- Hence the answer is 10815.

Questions for you to ponder

1. Why does this shortcut work?
2. How should we modify the procedure for the case when the two numbers are on opposite sides of the same power of 10? (For example, consider the product 97×102 .)

Finding the product of $999 \dots 99$ and a digit

We wish to compute the product $d \times 999 \dots 99$, where d is a digit ($d > 1$). Here is the procedure.

- Compute $a = d - 1$.
- Compute $B = 999 \dots 99 - a$.
- The answer is obtained by 'joining up' the two strings: aB .

Example: To compute 7×9999

- Here $d = 7$, so $a = d - 1 = 6$
- $9999 - 6 = 9993$.
- Hence the answer is 69993.

Example: To compute 8×99999

- Here $d = 8$, so $a = d - 1 = 7$
- $99999 - 7 = 99992$.
- Hence the answer is 799992.

Questions for you to ponder

1. Why does this shortcut work?
2. How should we modify the procedure for the case when $d > 9$ (i.e., d is not a digit)? (For example, consider the product 12×9999 .)

Explanations of earlier procedures

The two algorithms we consider below were described in the October 2024 issue. Please refer to the earlier issue for a description of the procedures.

Multiplying a number by 5: Explanation of the procedure

We consider separately the cases when n is even or odd, i.e., $n = 2m$ or $n = 2m + 1$ where m is a positive integer.

- If $n = 2m$ then $5n = 10m$, which is written as $\overline{m0}$.
- If $n = 2m + 1$, then $5n = 10m + 5$, which is just $\overline{m0} + 5 = \overline{m5}$.

This explains the procedure that we had described.

Squaring a number whose units digit is 5: Explanation of the procedure

A number whose units digit is 5 can be written $10A + 5$. Squaring this we get

$$\begin{aligned}(10A + 5)^2 &= 100A^2 + 100A + 25 \\ &= 100A(A + 1) + 25 = 100B + 25,\end{aligned}$$

where $B = A(A + 1)$. But $100B + 25$ is simply the number $\overline{B25}$. This explains the procedure that we had described.

Pretty Patterns – 4

In this piece, we display some visually striking patterns from nature. They have been taken from the article “26 Stunning Examples of Geometrical Symmetry Found in Nature” (URL: <https://121clicks.com/inspirations/geometrical-symmetry-found-in-nature>). The author writes,

Nature is a master artist, crafting patterns and shapes that exude balance and harmony. Geometrical symmetry, in particular, is a phenomenon that fascinates scientists, artists, and nature enthusiasts alike. From the microscopic world to vast landscapes, symmetry is a testament to the underlying order of the natural world. It's evident in the petals of flowers, the structure of snowflakes, and even the arrangement of galaxies, reminding us of nature's intrinsic connection to mathematics and beauty.

We can only marvel at the process that creates patterns of such beauty. Let us thank the author for assembling and sharing such wonderful pictures.



Figure 1: "Nature, This Is Some Nice Symmetry." Source: <https://imgur.com/gallery/nature-this-is-some-nice-symmetry-f35oi>



Figure 2: "Perfect Flower I Came Across While Hiking." Source: https://old.reddit.com/r/pics/comments/8kfp5f/perfect_flow_er_i_came_across_while_hiking/



Figure 3: "The Symmetry On Aloe Polyphylla Is Crazy." Source: https://old.reddit.com/r/succulents/comments/7srj2o/aloe_polyphylla_at_the_nursery_not_for_sale/



Figure 4: "This Nearly Perfect Dahlia I Spotted At A Local Park." Source: https://old.reddit.com/r/oddlysatisfying/comments/ovd3x5/this_nearly_perfect_dahlia_i_spotted_at_a_local/

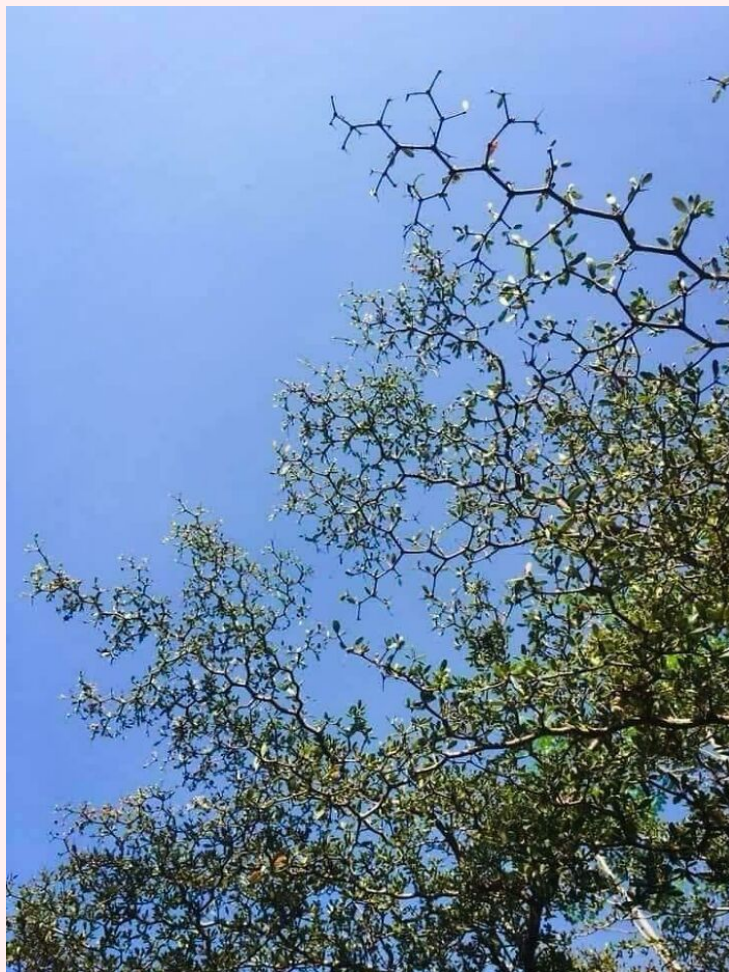


Figure 5: "Chemistree." Source: <https://bit.ly/3LwMPTf>

Bamboo Baskets and Math - Part 2

Article contributed by Padmapriya Shirali

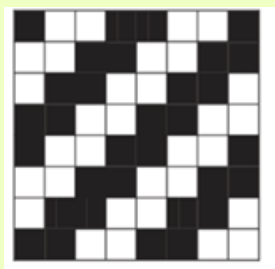
In Part 1 of this article (on “Bamboo Baskets and Math”) we looked at some basic designs seen in basket weaving, and how a repeating unit is used in each of the weaves. We explored ways to generate interesting shapes and patterns using three strips and four strips, respectively. Many familiar geometric concepts can be observed in these patterns — shapes, angles, symmetry, parallel lines, tessellations and slope.

Now, in Part 2 of the article, we will look at two types of twill patterns and explore ways of generating such patterns using basic arithmetic knowledge of the primary students.

"TWILL is a type of textile weave with a pattern of parallel, diagonal ribs. It is one of three fundamental types of weave, along with plain weave and satin. It is made by passing the weft thread over one or more warp threads then under two or more warp threads and so on, with a 'step', or offset, between rows to create the characteristic diagonal pattern. Because of this structure, twill generally drapes well." (This quote is from Wikipedia, "Twill." From <https://en.wikipedia.org/wiki/Twill>)

Twill fabrics, such as denim or tweed, are durable, flexible, and drape well. The diagonal texture distinguishes twill from plain weaves, making it ideal for clothing and upholstery.

Shown below is a twill 2-2 pattern that has a stagger of one vertical strip (Y axis / warp) in each row.



(a)



(b)

Figure 6: A twill 2-2 pattern

The first figure here is a grid drawing of a twill 2-2 pattern followed by an example of a woven cloth piece in the same pattern. We notice diagonal lines in this design that have a rib like appearance.

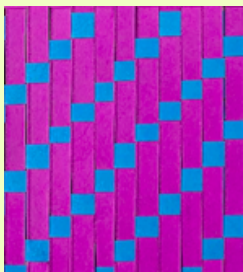
The code for this design would be as follows.

Row 1	1	2	2	2	1
Row 2	2	2	2	2	2

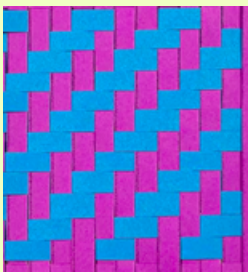
Table 1

Row 1 and Row 2 together are repeated in alternate colours.

Here are two more examples of a twill pattern with different coding rules.



(a)



(b)

Figure 7: From <https://handwovenmagazine.com/paper-weaving-intriguing-and-inspiring/>

Twill patterns can also be generated by students using the patterns in a multiplication table found in a number chart.

Here is a familiar chart showing numbers from 1 to 100, the numbers being arranged in columns of 10. Multiples of 3 have been highlighted. It produces a diagonal pattern.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

Figure 8

Children can investigate whether multiples of 3 will produce a different type of pattern if the numbers are arranged in columns of 8 (instead of 10).

Interestingly, that too produces a diagonal pattern — but it is in the opposite direction!

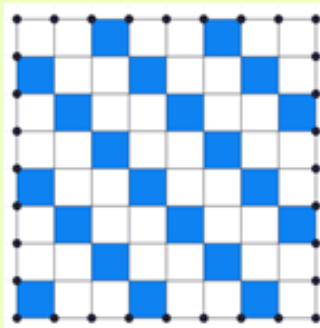


Figure 9

What happens if the numbers are arranged in 9 columns?

Is it possible to generate a pattern other than a straight or diagonal line with multiples of 3?

Would the line of squares on a 5-columns chart look the same as the line on a 10-column chart?

Would the line of squares on a 7-columns chart look the same as the line on a 10-column chart?

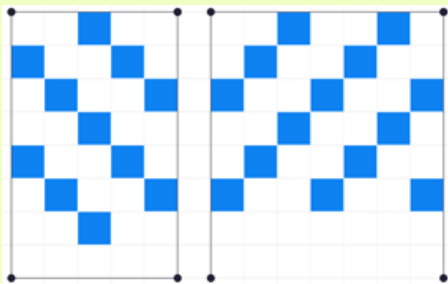


Figure 10

Is it possible to make a conjecture about the column numbers that 'look' the same (i.e., point in the same direction)?

Students can experiment with a few more numbers to notice the relationship between the number of columns and the directions of the lines of squares.

Students can also be given some patterns generated by multiples of other numbers.

Which multiples have generated this pattern?

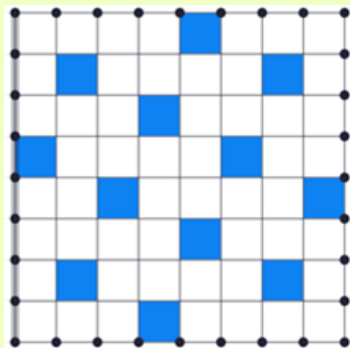


Figure 11

How would those multiples look on a 7-column grid?

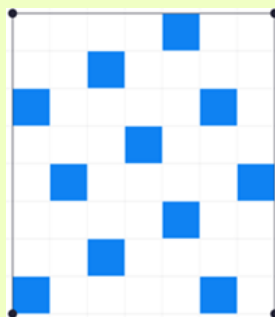


Figure 12

How would those multiples look on a 12-column grid?



Figure 13

Again, is there a relationship between the number of columns and the types of designs that can be generated?

Which column numbers will produce the same type of design?

Zig-zag twill

Now, we will look at the second type of pointed twill pattern called a *zig-zag twill*. In this pattern you will notice that the diagonal line doesn't progress continuously in one direction but reverses.

We will also see a point where the two twill lines meet. Herring bone weave is one example of a pointed twill.

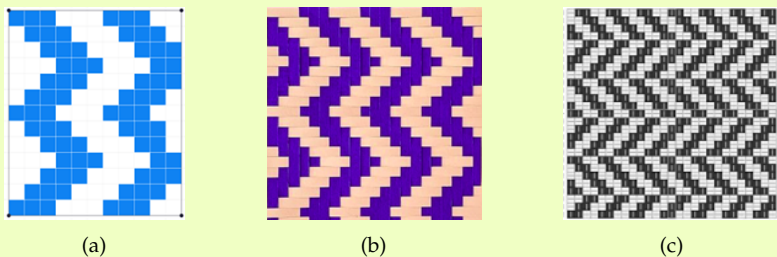


Figure 14

Figure (a) here is a grid drawing, followed by the paper weave, Figure (b), and an example of a cloth weave, Figure (c).

This follows a 1 2 3 4 3 2 1 pattern where 1 refers to the first row, 2 refers to the second row, etc.

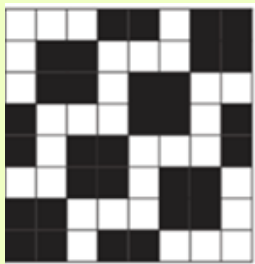
- Row 1: Over 3, Under 3... repeat
- Row 2: Under 1, then: Over 3, Under 3, ..., repeat
- Row 3: Under 2, then: Over 3, Under 3, ..., repeat
- Row 4: Under 3, Over 3, ..., repeat

Now the pattern is reversed. Row 5 is the same as Row 3, Row 6 is the same as Row 2, etc.

Students can be encouraged to try different numerical patterns that create zig-zag weaves. They could try with 1-2-3-2-1, or 1-2-3-4-5-4-3-2-1, and see what outcome results.

A few more patterns to try

To close, here are a few more interesting patterns for you to try.

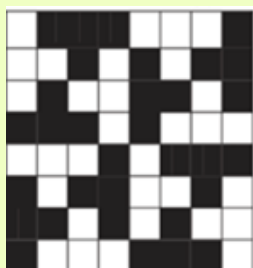


(a)

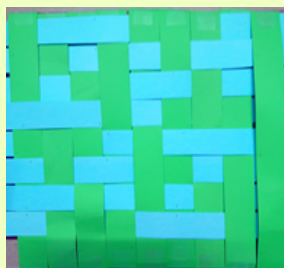


(b)

Figure 15: A hopsack pattern where the warp and weft create a simple checked pattern



(a)



(b)

Figure 16: A T pattern which shows movements of translation and rotational symmetry

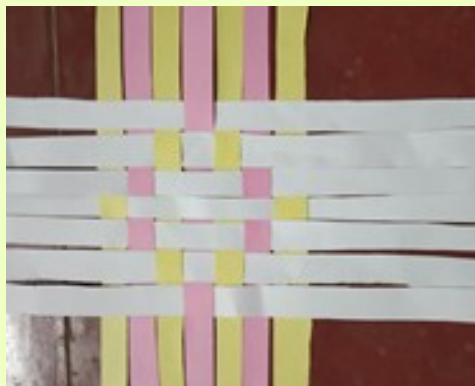


Figure 17: A Rhombus pattern

About the author

Padmapriya Shirali has been a math educator for the past four decades, teaching at various levels from Class 1 to Class 10. She has a strong interest in math pedagogy and conducting explorations in maths. She is a member of the NCERT textbook committee. She has worked closely with Shri P.K. Srinivasan, an eminent math educationist, and has been part of the Rishi Valley Rural Education team which developed the multi-grade learning material known as 'school in a box.' She may be contacted at padmapriya.shirali@gmail.com.

Playing with Primitive Pythagorean Triples – 4

Article contributed by Krittika Shiva

We continue our exploration of Primitive Pythagorean Triples (PPTs). In this issue we discuss yet another way of generating PPTs. It is an amazingly ‘low-tech’ approach! We also mention a striking divisibility property of PPTs and a result about the incircle of a Pythagorean triangle.

A ‘low-tech’ approach to generating PPTs

We are going to use the notion of simultaneous linear equations to generate PPTs. This should be a familiar topic; simultaneous linear equations are studied at the upper primary and high school levels. (That is why we have referred to this approach as a low-tech approach.)

We need to find a way of generating triples (a, b, c) of coprime positive

integers that solve the equation

$$a^2 + b^2 = c^2. \quad (1)$$

Divide through by c^2 and write

$$x = \frac{a}{c}, \quad y = \frac{b}{c}. \quad (2)$$

Then x, y are positive rational numbers ($0 < x, y < 1$) that solve the equation

$$x^2 + y^2 = 1. \quad (3)$$

We rewrite the above equation as follows: $y^2 = 1 - x^2 = (1 - x)(1 + x)$, hence

$$\frac{y}{1 - x} = \frac{1 + x}{y}. \quad (4)$$

Each fraction in the above equation is the ratio of two positive, rational numbers and is therefore a positive, rational number. Call this number t . Then we have

$$y = t(1 - x),$$

$$ty = 1 + x.$$

Rewrite the equations in the standard “simultaneous equations” format:

$$\begin{cases} tx + y &= t, \\ -x + ty &= 1. \end{cases} \quad (5)$$

Let us treat these as a pair of simultaneous linear equations in the two unknowns x, y and solve for these unknowns in terms of t , which is treated as a parameter.

Multiplying the second equation by t and adding, we get,

$$\begin{cases} tx + y = t, \\ -tx + t^2y = t, \end{cases} \quad \therefore (1+t^2)y = 2t,$$

hence,

$$y = \frac{2t}{1+t^2}.$$

From the equation $tx + y = t$ we get $x = 1 - y/t$, hence

$$x = 1 - \frac{2}{1+t^2} = \frac{1-t^2}{1+t^2}.$$

We have succeeded in expressing both x and y in terms of the parameter t :

$$x = \frac{1-t^2}{1+t^2}, \quad y = \frac{2t}{1+t^2}. \quad (6)$$

This representation is sufficient for generating all possible PPTs! All we need to do is substitute any positive rational value for t (any value will do, but, of course, it should be in its lowest terms), find the corresponding values of x and y , and then clear the fractions; and a PPT emerges from the dust of the battle! We illustrate this with some number crunching.

- Let $t = \frac{1}{2}$. This yields

$$x = \frac{1-1/4}{1+1/3} = \frac{3}{5}, \quad y = \frac{1}{1+1/4} = \frac{4}{5}.$$

Multiplying through by 5, we obtain the PPT (3,4,5).

- Let $t = \frac{2}{3}$. This yields

$$x = \frac{1-4/9}{1+4/9} = \frac{5}{13}, \quad y = \frac{4/3}{1+4/9} = \frac{12}{13}.$$

Multiplying through by 13, we obtain the PPT (5,12,13).

- Let $t = \frac{3}{4}$. This yields

$$x = \frac{1 - 9/16}{1 + 9/16} = \frac{7}{25}, \quad y = \frac{3/2}{1 + 9/16} = \frac{24}{25}.$$

Multiplying through by 25, we obtain the PPT (7,24,25).

- Let $t = \frac{2}{5}$. This yields

$$x = \frac{1 - 4/25}{1 + 4/25} = \frac{21}{29}, \quad y = \frac{4/5}{1 + 4/25} = \frac{20}{29}.$$

Multiplying through by 29, we obtain the PPT (20,21,29).

A puzzle

Observe the following.

- Let $t = \frac{1}{3}$. This yields

$$x = \frac{1 - 1/9}{1 + 1/9} = \frac{4}{5}, \quad y = \frac{2/3}{1 + 1/9} = \frac{3}{5}.$$

Multiplying through by 5, we obtain the PPT (4,3,5). We have obtained a permuted form of the PPT that we got using $t = \frac{1}{2}$.

- Let $t = \frac{1}{5}$. This yields

$$x = \frac{1 - 1/25}{1 + 1/25} = \frac{12}{13}, \quad y = \frac{2/5}{1 + 1/25} = \frac{5}{13}.$$

Multiplying through by 13, we obtain the PPT (12,5,13). This time we have obtained a permuted form of the PPT that we got using $t = \frac{2}{3}$.

We similarly find the following.

- The value $t = \frac{1}{4}$ yields $x = \frac{15}{17}$ and $y = \frac{8}{17}$. We get the PPT (15,8,17).

- The value $t = \frac{3}{5}$ yields $x = \frac{8}{17}$ and $y = \frac{15}{17}$. We get the PPT (8,15,17).
- The value $t = \frac{3}{4}$ yields $x = \frac{7}{25}$ and $y = \frac{24}{25}$. We get the PPT (7,24,25).
- The value $t = \frac{1}{7}$ yields $x = \frac{24}{25}$ and $y = \frac{7}{25}$. We get the PPT (24,7,25).

We see a very curious puzzle at work here; certain pairs of t -values give permuted forms of the same PPT! —

- $t = \frac{1}{2}$ and $t = \frac{1}{3}$
- $t = \frac{1}{5}$ and $t = \frac{2}{3}$
- $t = \frac{1}{4}$ and $t = \frac{3}{5}$
- $t = \frac{3}{4}$ and $t = \frac{1}{7}$.

What's the underlying pattern here? What law connects the two values of t that yield permutations of the same PPT? We leave the unravelling of this puzzle to the reader. You can add to the database of pairs of related t -values by doing some computations yourself.

A divisibility property of PPTs

Here is an interesting property which you may have noticed when you looked at a list of PPTs.

To lead up to the property, we list a few PPTs here and form the products of the three numbers in each triple.

- PPT: (3,4,5); product: $2^2 \cdot 3 \cdot 5 = 60$.
- PPT: (5,12,13); product: $2^2 \cdot 3 \cdot 5 \cdot 13 = 780$.
- PPT: (7,24,25); product: $2^3 \cdot 3 \cdot 5^2 \cdot 7 = 4200$.

- PPT: (20,21,29); product: $2^2 \cdot 3 \cdot 5 \cdot 7 \cdot 29 = 12180$.
- PPT: (8,15,17); product: $2^3 \cdot 3 \cdot 5 \cdot 17 = 2040$.

We observe immediately that all the products share the following feature: the presence of the factors 2^2 , 3 and 5 in the prime factorization of the product. We state this in a more striking form as follows.

Theorem. *The product of the three numbers in a PPT is a multiple of 60.*

This is implied by the following:

Theorem. *In any PPT, one of the numbers is a multiple of 3, one of the numbers is a multiple of 4, and one of the numbers is a multiple of 5.*

It should be clear that if this is true, then the product of the numbers in a PPT will be a multiple of 60.

To prove this, we use the following facts.

- (a) *All square numbers are $0 \pmod{3}$ or $1 \pmod{3}$.*

This follows from the following (easy to check) congruences modulo 3:

$$0^2 \equiv 0, \quad 1^2 \equiv 1, \quad 2^2 \equiv 1.$$

So, there are no square numbers of the form $2 \pmod{3}$

- (b) *All square numbers are $0 \pmod{4}$ or $1 \pmod{4}$.*

This follows from the following (easy to check) congruences modulo 4:

$$0^2 \equiv 0, \quad 1^2 \equiv 1, \quad 2^2 \equiv 0, \quad 3^2 \equiv 1.$$

So, there are no square numbers of the forms $2 \pmod{4}$ or $3 \pmod{4}$.

(c) All *odd square numbers* are $1 \pmod{8}$.

This follows from the following (easy to check) congruences modulo 8:

$$1^2 \equiv 1, \quad 3^2 = 9 \equiv 1, \quad 5^2 = 25 \equiv 1, \quad 7^2 = 49 \equiv 1.$$

(d) All *square numbers* are $0 \pmod{5}$ or $1 \pmod{5}$ or $-1 \pmod{5}$.

This follows from the following (easy to check) congruences modulo 5:

$$0^2 \equiv 0, \quad 1^2 \equiv 1, \quad 2^2 \equiv 4 \equiv -1, \quad 3^2 \equiv 4 \equiv -1, \quad 4^2 \equiv 1.$$

So, there are no square numbers of the forms $2 \pmod{5}$ or $3 \pmod{5}$.

We now prove the main proposition. Let (a, b, c) be a PPT. We shall prove the following.

(i) *One of a, b, c is a multiple of 3.*

For, suppose that this is not the case. That is, suppose that a, b, c are all indivisible by 3. Then we have

$$a^2 \equiv 1 \pmod{3}, \quad b^2 \equiv 1 \pmod{3}, \quad c^2 \equiv 1 \pmod{3}.$$

This contradicts the equality $a^2 + b^2 = c^2$, since $1 + 1 \not\equiv 1 \pmod{3}$. Hence the conclusion.

(ii) *One of a, b, c is a multiple of 4.*

For, it cannot be that all three of a, b, c are odd (we would then have $\text{Odd} + \text{Odd} = \text{Even}$ on one side and Odd on the other side). It also cannot be that just two of a, b, c are even (for similar reasons). So, just one of a, b, c is even and the other two are odd.

Can it be that c is even and a, b are odd? If so, we would have

$$a^2 \equiv 1 \pmod{4}, \quad b^2 \equiv 1 \pmod{4}, \quad c^2 \equiv 0 \pmod{4}.$$

This contradicts the equality $a^2 + b^2 = c^2$, since $1 + 1 \not\equiv 0 \pmod{4}$. So c is odd, and one of a, b is odd while the other is even.

Without any loss of generality we may suppose that a is even and b is odd. Then we have

$$b^2 \equiv 1 \pmod{8}, \quad c^2 \equiv 1 \pmod{8},$$

hence $a^2 \equiv 0 \pmod{8}$. So, a^2 is divisible by 8, which implies that a is divisible by 4. (It cannot be that a is an even number not divisible by 4, as that would lead to $a^2 \equiv 4 \pmod{8}$.) Hence the conclusion.

(iii) *One of a, b, c is a multiple of 5.*

For, suppose that this is not the case. That is, suppose that a, b, c are all indivisible by 5. Then we have

$$a^2 \equiv \pm 1 \pmod{5}, \quad b^2 \equiv \pm 1 \pmod{5}, \quad c^2 \equiv \pm 1 \pmod{5}.$$

This contradicts the equality $a^2 + b^2 = c^2$, since $\pm 1 \pm 1 \not\equiv \pm 1 \pmod{5}$ for any combination of $\pm$ signs. Hence the conclusion.

Though we have not explicitly mentioned it, more can be said:

- Exactly one of a, b, c is a multiple of 3;
- Exactly one of a, b, c is a multiple of 4;
- Exactly one of a, b, c is a multiple of 5.

You should prove these statements for yourself.

It follows from the above that abc is divisible by $3 \times 4 \times 5 = 60$. ■

Remark 1 We stated above that exactly one of a, b, c is a multiple of 3, exactly one of a, b, c is a multiple of 4, and exactly one of a, b, c is a multiple of 5. Note that it is possible for the same number in the triple to be a multiple of 3, of 4, and of 5. This happens, for example, with the PPT (11, 60, 61).

Remark 2 The above result cannot be ‘improved,’ in the sense that the number 60 cannot be replaced by any larger number. It is easy to see why: the PPT (3, 4, 5) has product exactly 60.

About the author

Krittika Shiva is a volunteer at the Community Mathematics Centre and helps out with the editing work for ‘The Compass.’ She has a deep interest in elementary number theory and its history. She may be contacted at compass.math.2024@gmail.com.

The Big Wide World of Data

Visualisation – 4

As stated in earlier issues, this column aims to present a running series of short articles on the theme of data visualisation. We shall present a few striking examples of this in each issue (with little or no commentary).

Time spent watching videos on social media

From the article “We’re spending more time watching videos on social media” by Millie Giles & David Crowther, available at https://sherwood.news/culture/social-media-increasingly-just-social-video/?utm_source=chartr&utm_medium=email&utm_campaign=chartr_20240719

Streaming vs. Social media

For some time, small screens and big screens have been competing for our attention, with many people admitting to “second screening” – when they find themselves flicking through their phone while also absent-mindedly watching something on TV. Now, a new report from Variety suggests that a rise in video consumption on social media is eating into the time we spend streaming on-demand ... and TikTok is leading the charge.

Indeed, the average daily video watch time has increased from 2.12 hours in 2022 to 2.48 hours this year on TikTok, according to data from Media IDentity Graph supplied to Variety. That’s more than any other major social media platform’s watch time, many of which have pivoted to video to compete with TikTok (as with Instagram’s Reels feature).

With audiences spending more time on social video – video now accounts for almost 60% of average time spent on social networks, according to eMarketer – TikTok is now testing a 60-min length limit for uploads, upping the 15-min max established last year, and considerably greater than the 15-second ceiling that it had at launch.

Now, short- and long-form social video is cutting into time spent watching streaming, particularly for younger consumers: per Variety’s report, “Watching TV and movies together accounted for just 32% of media time among 13-24 year-olds, versus 59% for consumers over 35.”

In fact, 58% of respondents aged 13-24 reported that they were spending less time watching “regular” TV because they were watching “non-premium” online videos (i.e., social video). Just as cable TV was quickly swallowed by on-demand streaming at the turn of the century, it was inevitable that an even bigger fish – and, in this case, a smaller screen –

would eventually come along.

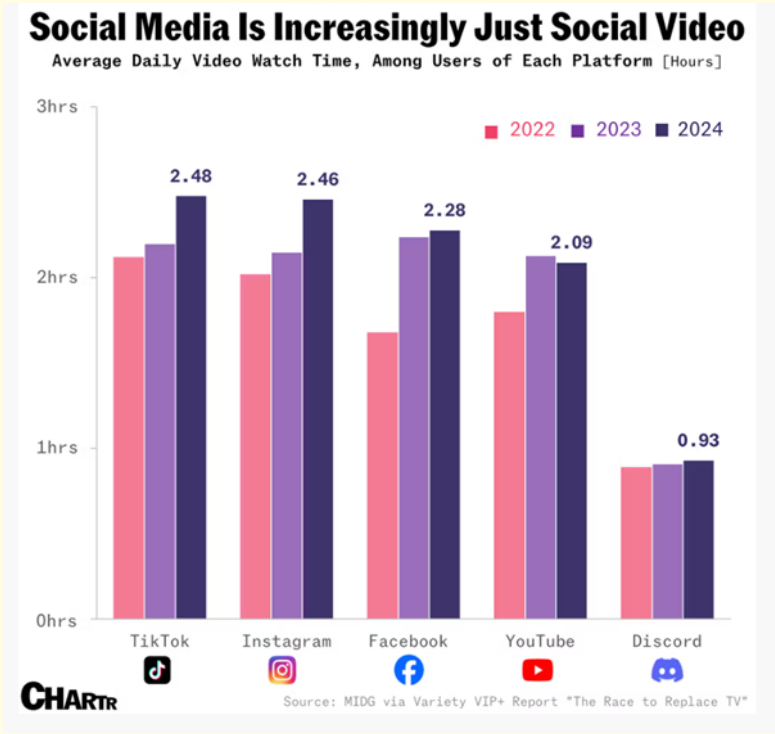


Figure 18: Streaming vs. Social media. Source: https://sherwood.news/culture/social-media-increasingly-just-social-video/?utm_source=chartr&utm_medium=email&utm_campaign=chartr_20240719

Wikipedia Searches Reveal Differing Styles of Curiosity

From the article “Wikipedia Searches Reveal Differing Styles of Curiosity” by Gary Stix, available at <https://www.scientificamerican.com/article/wikipedia-searches-reveal-differing-styles-of-curiosity/>

A recent study using Wikipedia has identified three distinct styles of human curiosity: the *busybody*, the *hunter*, and the *dancer*. These styles were uncovered by tracking how over 482,000 users across 50 countries and 14 languages navigated Wikipedia’s network of interconnected articles. This research, published in **Science Advances**, highlights the nuanced ways people seek and connect information, offering new insights into the nature of curiosity.

Busybody users explore numerous, often unrelated topics in a zigzagging path, showcasing a broad but scattered curiosity. Hunters, in contrast, focus deeply on a smaller set of closely related topics, exhibiting targeted inquisitiveness. Dancers combine highly disparate topics, aiming to synthesize new ideas. Dani Bassett, a co-senior author of the study and a network scientist at the University of Pennsylvania, explained that curiosity is not just about collecting information but connecting it to what we already know.

The researchers used “knowledge networks” to map how search topics are interrelated, linking curiosity styles to variables such as education, gender equality, and well-being. They found that

in countries with higher education levels and greater gender equality, users tended to browse more like busybodies. Conversely, in places with lower levels of education and gender equality, people browsed more like hunters. Bassett hypothesizes that societal structures, such as patriarchal constraints, may limit the freedom to explore broadly, encouraging more focused, goal-oriented behaviour.

This study integrates expertise from diverse fields, including psychology, sociology, philosophy, and computational modelling, to provide a comprehensive understanding of human inquisitiveness. It builds on philosophical concepts of curiosity developed by thinkers like Heidegger and Nietzsche, which were explored by Bassett's twin brother, Perry Zurn, a philosopher at American University. Zurn's analysis of 2,000 years of Western thought inspired the busybody-hunter-dancer framework, which was then tested using Wikipedia's vast, real-world dataset.

Princeton psychologist Erik Nook praised the study's interdisciplinary approach and its immense scope, calling it a "dazzlingly large" effort to understand human behaviour. The work also reveals broader implications for how social structures influence curiosity and knowledge-seeking behaviours. By examining these curiosity styles and their connections to societal factors, the study provides a deeper understanding of how humans explore, learn, and innovate in a complex, interconnected world.

How Different Styles of Curiosity Drive Wikipedia Searches

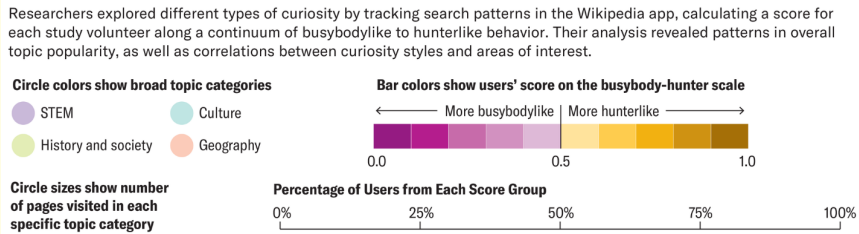


Figure 19: Wikipedia Searches Reveal Differing Styles of Curiosity. Source: <https://www.scientificamerican.com/article/wikipedia-searches-reveal-differing-styles-of-curiosity/>

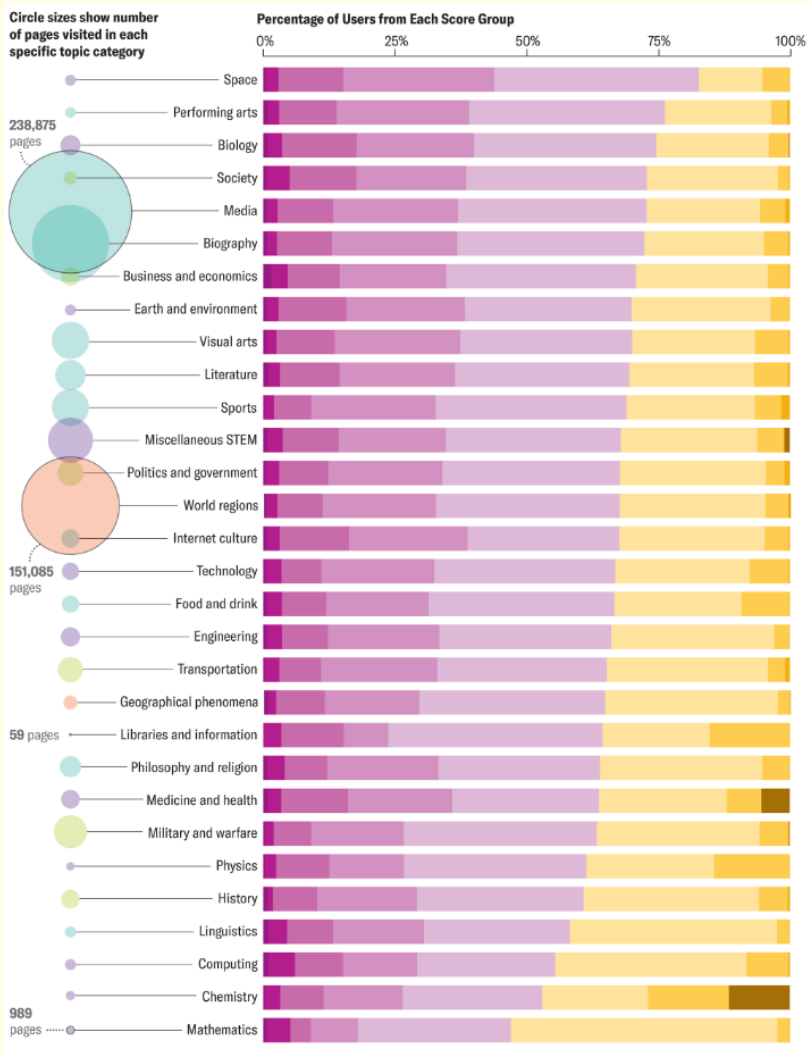


Figure 20: Wikipedia Searches Reveal Differing Styles of Curiosity

2024 Is Officially the Hottest Year on Record

Earth Surpasses 1.5 Degrees C in Hottest Year on Record

The year 2024 was the hottest on record and the first to top 1.5 degrees Celsius of warming. All 10 of the hottest years have been in the past decade

From the article “2024 Is Officially the Hottest Year on Record” by Andrea Thompson, available at <https://www.scientificamerican.com/article/2024-is-officially-the-hottest-year-on-record/>

The year 2024 has officially become the hottest year on record, marking the first time global temperatures exceeded 1.5°C above pre-industrial levels. This milestone underscores how much the climate has shifted due to the continued burning of fossil fuels. According to the European Union’s Copernicus Climate Change Service (C3S), 2024 surpassed 2023, which had already broken the previous record by a considerable margin. All of the 10 hottest years on record have occurred in the past decade, highlighting the acceleration of global warming.

The Paris climate agreement aims to limit warming to below 1.5°C and well under 2°C, but this threshold is now precariously close. Samantha Burgess, a strategic lead at the European Center for Medium-Range Weather Forecasts, noted that the average temperature over the past two years has already exceeded 1.5°C. While natural climate phenomena like El Niño have contributed to recent temperature spikes by releasing heat from the Pacific Ocean into the atmosphere, the primary driver of the current warming trend is the accumulation of greenhouse gases.

Carbon dioxide (CO₂) levels, the leading contributor to global warming, reached a record high of 420 parts per million (ppm) in 2023, compared to 280 ppm during pre-industrial times. This excess heat has fueled catastrophic climate impacts, such as wildfires in Los Angeles, emphasizing the urgent need for rapid and substantial global emissions reductions. Carlo Buontempo, director of C3S, stressed that the future depends on swift and decisive action to alter the trajectory of climate change.

Despite the rapid growth of renewable energy, rising energy demand has offset significant progress in reducing emissions. Political challenges, such as President-elect Donald Trump's commitment to expanding fossil fuel production and weakening emissions regulations, further complicate efforts to combat climate change.

Looking ahead, experts do not expect 2025 to surpass 2024's record heat, partly due to the onset of La Niña. This phenomenon, characterized by cooler Pacific Ocean waters, is expected to slightly cool global temperatures. However, this cooling effect will be minimal, and 2025 is still predicted to rank among the three hottest years on record. Adam Scaife of the U.K.'s Met Office remarked that 2016, previously the warmest year due to El Niño, now appears "decidedly cool" compared to recent projections. The escalating trend underscores the urgent need for global climate action.

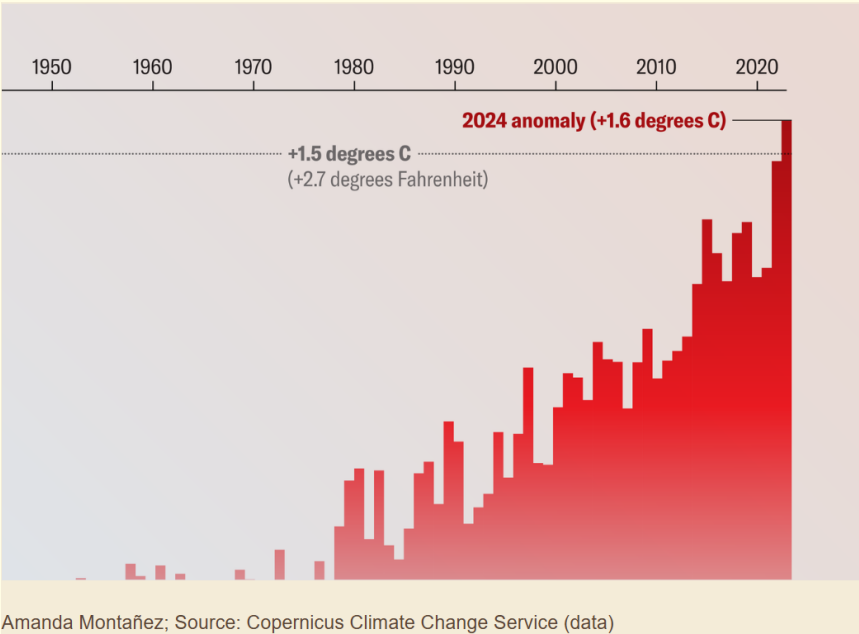


Figure 21: Source: <https://www.scientificamerican.com/article/2024-is-officially-the-hottest-year-on-record/>

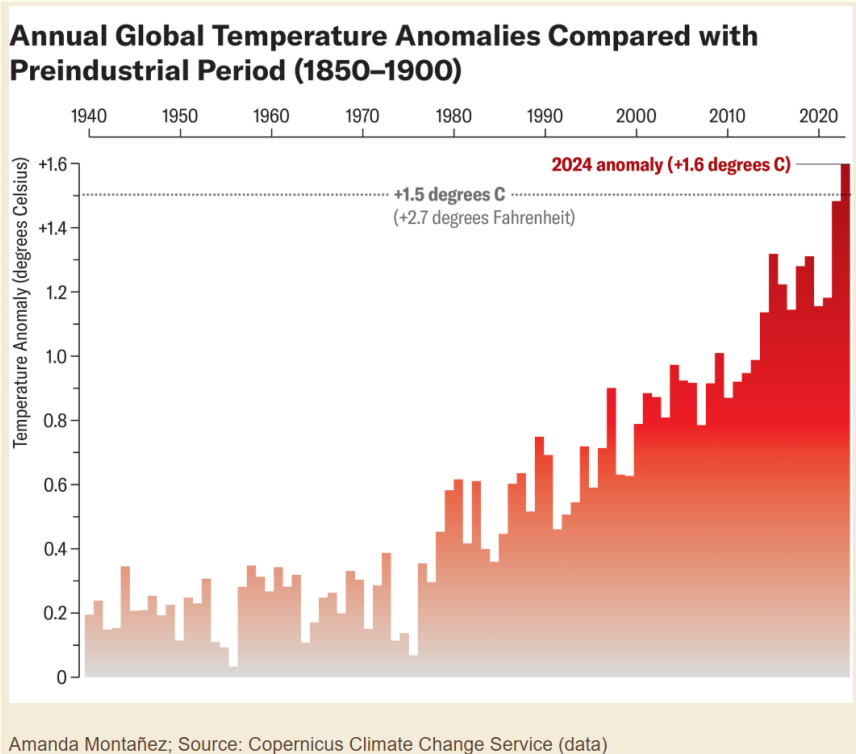


Figure 22: Source: <https://www.scientificamerican.com/article/2024-is-officially-the-hottest-year-on-record/>

How To Prove It – 4

We continue our series of articles on the theme of proof. In previous issues we studied *proof by induction* and *proof by enumeration of all possibilities* (also known informally as brute force enumeration). In this issue we consider the important notion of a *counterexample*.

Notion of a counterexample

A ‘counterexample’ is simply a specific instance that contradicts (and therefore disproves) a claim of a general nature.

To illustrate what this means consider the following often-quoted sentence from Albert Einstein (from *The New York Times*, 29 March 1972, and *The New York Post*, 28 November 1972).

No amount of experimentation can ever prove me right; a single experiment can prove me wrong.

This compact sentence captures the central spirit of the scientific method. With each successful experiment (‘successful’ in the sense that its results

confirm the predictions made before conducting the experiment), evidence accumulates that the underlying theory is correct. But a single experiment in which the results are contrary to the predictions made is enough to destroy the theory. Such an outcome is known as a *counterexample*.

In passing we share the following apocryphal story of the early 1930s when vigorous attempts were being made in Nazi Germany to discredit Einstein. The story goes that Einstein saw a headline in a German newspaper stating, “One hundred German physicists claim Einstein’s theory of relativity is wrong.” He is reported to have remarked, “If I were wrong, it would only take one.”

Einstein’s quote not only captures the essence of the scientific method (the requirement that a scientific theory must in principle be falsifiable) but also illustrates a central aspect of mathematics. When a claim is made in mathematics, one may compile a hundred instances that are in perfect agreement with the claim, but this has no value at all. It may well happen that the very next experiment turns up a contradictory outcome. If this happens, the claim is destroyed for ever. This may seem a depressing outcome, but in fact disproving a claim is just as vital to progress in mathematics as proving a claim!

We describe in this article a few instances (mostly about prime numbers) of such a contradictory outcome. The first two examples are trivial in nature but have been included mainly to illustrate the theme.

Example 1: Industrial grade primes

Claim: “All odd numbers greater than 1 are prime.”

Let’s check: 3, 5 and 7 certainly are prime, but (sadly) the very next odd

number fails the test: 9 is not prime (so it is a counterexample). Hence, the claim is false.

Example 2: Divisibility by 2, 3, 5

Claim: “If a number is not divisible by 2 or 3 or 5, then it is prime.”
(Expressed differently: “If a number is coprime to 30, then it is prime.”)

Finding a counterexample takes more work than earlier, but eventually we do find one. The smallest counterexample is the number $49 = 7 \times 7$. It is coprime to 30, but it is not prime. Hence, the claim is false.

Example 3: Mersenne numbers

Claim: “If n is prime, then $2^n - 1$ is prime.”

Let’s check. The first few primes are 2, 3, 5, 7. The stated claim is true for these primes, since $2^2 - 1 = 3$, $2^3 - 1 = 7$, $2^5 - 1 = 31$, $2^7 - 1 = 127$ are all prime. But the next trial pulls down the claim, since

$$2^{11} - 1 = 2047 = 23 \times 89.$$

We urge the reader to check what happens for $n = 13, 17, 19$ and 23.

Remark Numbers of the form are known as *Mersenne numbers*, and prime numbers of this form are known as *Mersenne primes*. It is easy to prove the following:

Theorem. *If n is a positive integer such that $2^n - 1$ is prime, then n is prime.*
More compactly:

$$2^n - 1 \text{ PRIME} \implies n \text{ PRIME}.$$

An equivalent way of stating the above is the following (and in this form it is very easy to prove):

Theorem. *If n is composite, then $2^n - 1$ is composite. More compactly:*

$$n \text{ COMPOSITE} \implies 2^n - 1 \text{ COMPOSITE.}$$

The reader should try proving this result on his or her own. (Hint: Let $n = ab$ where $a > 1$, $b > 1$. Prove that $2^{ab} - 1$ is divisible both by $2^a - 1$ and by $2^b - 1$. Conclude that $2^{ab} - 1$ is composite.)

Armed with this result, it is quite easy to jump to the conclusion that the converse statement is true; that is, $n \text{ PRIME} \implies 2^n - 1 \text{ PRIME}$. But as we noted above, this claim is false; indeed, $n = 11$ is a counterexample.

It is known (see Example 5, below) that testing arbitrary numbers for primality is a difficult task, computationally; most available tests consume too much time. But for the special case of Mersenne numbers, a test is available (the “Lucas–Lehmer primality test”, [2]) that can be executed efficiently. For this reason, at any given moment the largest known prime is most often a Mersenne prime. Right now, (March 2025) the largest known prime is the following number which has more than 41 million digits:

$$2^{136279841} - 1.$$

For more about these primes the reader should refer to the interested reader can refer to [1].

Example 4: Fermat numbers and Euler’s remarkable feat

Claim: “If n is any positive integer, then $2^{2^n} + 1$ is prime.”

This claim has a very interesting history as the statement was made by no less a mathematician than Pierre Fermat! Checking this claim in a pre-calculator or pre-computer age is daunting as the numbers involved become large very quickly.

For convenience, let us write $f(n) = 2^{2^n} + 1$. Then:

- $f(1) = 5$, which is prime.
- $f(2) = 17$, which is prime.
- $f(3) = 257$, which is prime.
- $f(4) = 65537$, which is prime.

Checking by hand whether $f(4)$ is prime is no easy task — but it can be done. So we may satisfy ourselves that Fermat's claim is true for $n = 1, 2, 3, 4$.

But what do we do with the next number in the series,

$$f(5) = 2^{32} + 1 = 4294967297.$$

Checking (by hand calculation) whether or not 4294967297 is prime may seem a hopeless task. But this seemingly impossible task was carried out by Leonard Euler, who showed that

$$4294967297 = 641 \times 6700417.$$

(Both numbers on the right side are primes!) How Euler performed this feat is a story in itself which we will narrate on some other occasion.

Example 5: A test for primality?

The reader will no doubt be familiar with various tests for divisibility — by 3, by 9, by 11, by 7, by 13, etc. It may come as a shock to hear that whereas

tests for divisibility can be found for any divisor whatsoever, there is no simple way of testing a number for primality; i.e., checking whether a given number is prime or not.

Many centuries back, it was believed (in some parts of the world) that the following is a test for primality.

Test for primality: *Let n be greater than 1. Check whether or not $2^n - 2$ is divisible by n . If it is divisible, then declare n to be prime. If it is not divisible, then declare n to be composite. In short, for odd numbers $n > 1$,*

$$n \text{ PRIME} \iff 2^n - 2 \text{ IS DIVISIBLE BY } n.$$

But is this a true statement? Let's check.

- Let $n = 7$ (a prime number). Is $2^7 - 2 = 126$ divisible by 7? It is; $126 \div 7 = 18$.
- Let $n = 11$ (a prime number). Is $2^{11} - 2 = 2046$ divisible by 11? It is; $2046 \div 11 = 186$.
- Let $n = 13$ (a prime number). Is $2^{13} - 2 = 8190$ divisible by 13? It is; $8190 \div 13 = 630$.
- Let $n = 9$ (a composite number). Is $2^9 - 2 = 510$ divisible by 9? It isn't.
- Let $n = 10$ (a composite number). Is $2^{10} - 2 = 1022$ divisible by 10? It isn't.
- Let $n = 15$ (a composite number). Is $2^{15} - 2 = 32766$ divisible by 15? It isn't.

Please try out the test for more values of n . You'll find yourself getting convinced fairly soon that the test is valid!

But you are in for a surprise. *The test isn't valid*. Finding a counterexample takes time but we do spot one eventually. The smallest counterexample turns out to be 341. This number is composite ($341 = 11 \times 31$), but $2^{341} - 2$ is divisible by 341.

Here's something for you to ponder. How will you verify that $2^{341} - 2$ is divisible by 341 without having to compute the number $2^{341} - 2$? (No one in a reasonable frame of mind would want to compute this number — it has 103 digits.)

Further reading

[1] Wikipedia, "Mersenne prime" from

https://en.wikipedia.org/wiki/Mersenne_prime

[2] Wikipedia, "Lucas-Lehmer primality test" from [https://en.wikipedia.org/wiki/Lucas-Lehmer_primality_test](https://en.wikipedia.org/wiki/Lucas%E2%80%93Lehmer_primality_test)

Article contributed by Shailesh Shirali

The Angle Bisector Theorem – Part 1

Article contributed by Rohini Chandra

This two-part article is devoted to a study of the Angle Bisector Theorem — a simple-looking result in triangle geometry of great utility.

In Part 1 we study the theorem and its proof. We also study the converse result.

In Part 2 we study some interesting applications of the theorem.

Here is its statement followed by two different proofs.

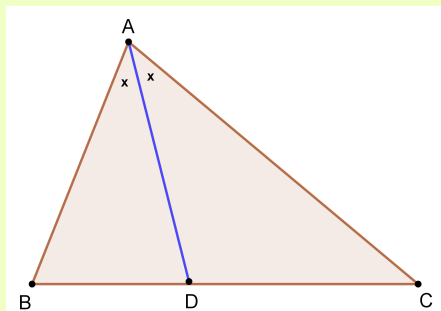


Figure 23: The angle bisector theorem

Here are two different proofs of the theorem.

First proof, using areas

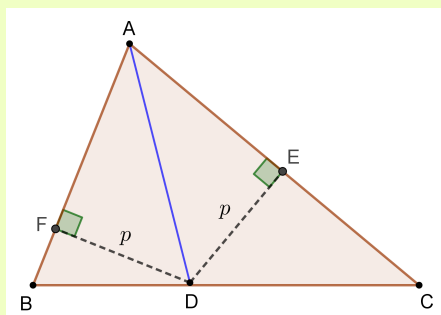


Figure 24: Area argument to prove the angle bisector theorem

Consider $\triangle ABD$ & $\triangle ACD$:

$$\frac{\text{Area of } \triangle ABD}{\text{Area of } \triangle ACD} = \frac{BD}{DC}, \quad (7)$$

since the two triangles have equal height (which is the distance from A to BC).

We also have,

$$\frac{\text{Area of } \triangle ABD}{\text{Area of } \triangle ACD} = \frac{\frac{1}{2}AB \cdot p}{\frac{1}{2}AC \cdot p}, \quad (8)$$

where p is the perpendicular distance from D to AB or AC (observe that the two distances are equal because AD is an angle bisector). Hence

$$\frac{\text{Area of } \triangle ABD}{\text{Area of } \triangle ACD} = \frac{AB}{AC}. \quad (9)$$

It follows that

$$\frac{AB}{AC} = \frac{BD}{DC}.$$

Comment

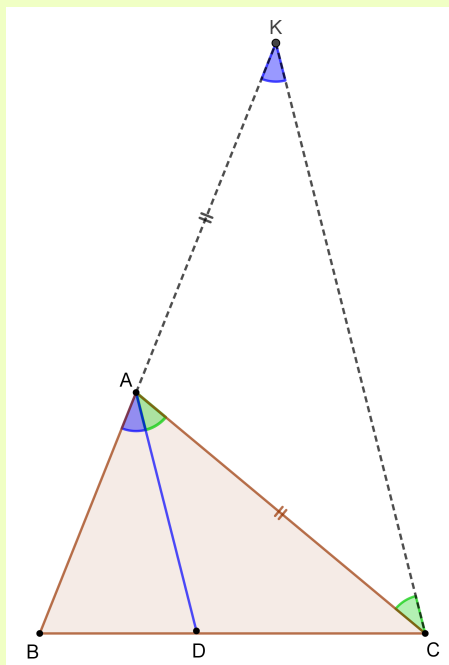
The equality (9) can also be arrived at using the sine formula for area:

$$\begin{aligned} \frac{\text{Area of } \triangle ABD}{\text{Area of } \triangle ACD} &= \frac{\frac{1}{2}AB \cdot AD \cdot \sin \angle BAD}{\frac{1}{2}AC \cdot AD \cdot \sin \angle CAD} \\ &= \frac{AB}{AC}, \end{aligned}$$

since $\angle BAD = \frac{1}{2}\angle A = \angle CAD$. ■

The second proof is based on the notion of similarity of triangles.

Second proof, using similar triangles



Construction: Extend BA beyond A to point K such that $AK = AC$.

Then we have

$$\angle AKC = \angle ACK = \frac{1}{2} \angle A,$$

hence $KC \parallel AD$. It follows that

$$\frac{BA}{AK} = \frac{BD}{DC},$$

and since $AK = AC$, the desired equality follows:

$$\frac{AB}{AC} = \frac{BD}{DC}.$$

Figure 25: Proof of the angle bisector theorem using similarity of triangles

Converse of the angle bisector theorem

The theorem stated above has an obvious converse:

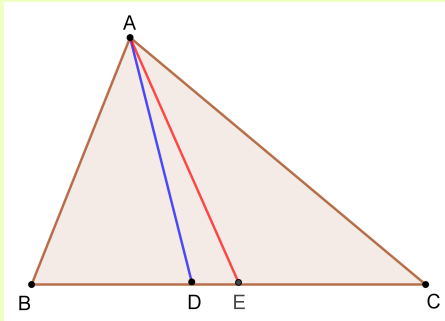
Theorem. In $\triangle ABC$, let D be a point on side BC such that

$$\frac{AB}{AC} = \frac{BD}{DC}.$$

Then AD is the internal bisector of $\angle BAC$.

Proof of the converse theorem

Here are two approaches by which we can prove the Converse Theorem.



Given: D is a point on side BC such that

$$\frac{AB}{AC} = \frac{BD}{DC}. \quad (10)$$

Let AE be the internal bisector of $\angle BAC$ (with E on BC).

Figure 26

We now invoke the angle bisector theorem. The theorem tells us that

$$\frac{AB}{AC} = \frac{BE}{EC}. \quad (11)$$

From (10) and (11) we see that

$$\begin{aligned} \frac{BE}{EC} &= \frac{BD}{DC}, \\ \therefore \frac{BE}{EC} + 1 &= \frac{BD}{DC} + 1, \\ \therefore \frac{BC}{EC} &= \frac{BC}{DC}, \end{aligned}$$

therefore $EC = DC$, which can only happen if D and E coincide.

It follows that AD is the internal bisector of $\angle BAC$, as required. ■

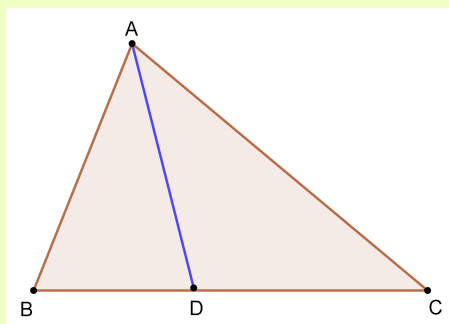
A trigonometric approach

Figure 27

Given: D is a point on side BC such that

$$\frac{AB}{AC} = \frac{BD}{DC}. \quad (12)$$

We now use the sine formula for area of a triangle.

Then we have,

$$\begin{aligned} \frac{BD}{DC} &= \frac{\text{Area of } \triangle ABD}{\text{Area of } \triangle ACD} = \frac{\frac{1}{2}AB \cdot AD \cdot \sin \angle BAD}{\frac{1}{2}AC \cdot AD \cdot \sin \angle DAC} \\ &= \frac{AB \cdot \sin \angle BAD}{AC \cdot \sin \angle DAC}. \end{aligned} \quad (13)$$

But we are told (12) that

$$\frac{AB}{AC} = \frac{BD}{DC}.$$

Hence we have

$$\sin \angle BAD = \sin \angle DAC.$$

We know that if two angles between 0° and 180° have equal sine, then they are either equal or supplementary (i.e., they add up to 180°).

The latter possibility would mean that $\angle BAC = 180^\circ$, which can be ruled out. (It would mean that points B, A, C are collinear.)

Hence $\angle BAD = \angle DAC$, i.e., AD is the internal bisector of $\angle BAC$, as required. 

In the second part of this article we study a few applications.

About the author

Rohini Chandra is a student of mathematics with a particular interest in Euclidean geometry and its history.

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Gems From Geometry – 3

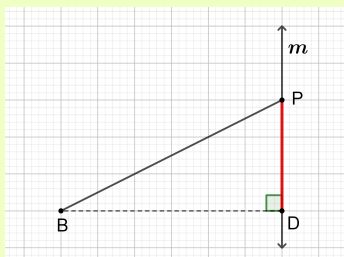
A cousin of Viviani's theorem

In the previous issue we studied Viviani's theorem — the statement that for a given equilateral triangle, the sum of the perpendicular distances from any interior point to the sides of the triangle is the same for all interior points. We offered different proofs of the proposition. Now, we study what we should call a “cousin” of Viviani's theorem.

The history of this result highlights important questions related to the teaching of mathematics. It was discovered empirically by Duncan Clough, a Cape Town (South Africa) grade 11 student, during a session in which the students were exploring Viviani's theorem (using dynamic geometry software) and attempting to prove it; he reported it to his teacher Marcus Bizony, who wrote to Michael de Villiers of the Department of Math Education, University of KwaZulu-Natal, South Africa; and that's how it got the name “Clough's Conjecture.” It is now a theorem, proved by de Villiers himself; see [1]. In the paper, the author notes how the search for proof sometimes uncovers new results. He shows how the theorem follows from

Viviani's theorem and deduces some extensions, e.g., to a rhombus and to an equiangular pentagon.

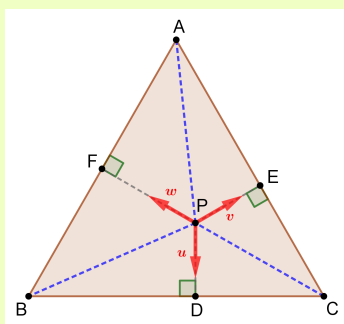
To introduce the 'cousin' we recall the vector proof of Viviani's theorem from the January 2025 issue. The proof drew on the following observation.



Let PB be a segment; let m be a line through P (Figure 28). Let PD be the projection of PB on m . Then $|PD| = |\vec{PB} \cdot \mathbf{u}|$, where $\mathbf{u}$ is a unit vector along the line m .

Figure 28: An observation about projections

We apply this to the proof of Viviani's theorem.



Let P be a point within $\triangle ABC$. Let $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$ be unit vectors perpendicular respectively to BC , CA , AB , as shown in Figure 29. Then the vector sum $\mathbf{u} + \mathbf{v} + \mathbf{w}$ is the zero vector. (This follows from symmetry.)

Figure 29

Drawing on the observation made above, we get,

$$PD = \overrightarrow{PB} \cdot \mathbf{u}, \quad PE = \overrightarrow{PC} \cdot \mathbf{v}, \quad PF = \overrightarrow{PA} \cdot \mathbf{w}.$$

For some other point Q within the triangle, we similarly have

$$QD = \overrightarrow{QB} \cdot \mathbf{u}, \quad QE = \overrightarrow{QC} \cdot \mathbf{v}, \quad QF = \overrightarrow{QA} \cdot \mathbf{w}.$$

By addition and subtraction, we get,

$$\begin{aligned} & (PD + PE + PF) - (QD + QE + QF) \\ &= (\overrightarrow{PB} \cdot \mathbf{u} + \overrightarrow{PC} \cdot \mathbf{v} + \overrightarrow{PA} \cdot \mathbf{w}) - (\overrightarrow{QB} \cdot \mathbf{u} + \overrightarrow{QC} \cdot \mathbf{v} + \overrightarrow{QA} \cdot \mathbf{w}) \\ &= (\overrightarrow{PB} - \overrightarrow{QB}) \cdot \mathbf{u} + (\overrightarrow{PC} - \overrightarrow{QC}) \cdot \mathbf{v} + (\overrightarrow{PA} - \overrightarrow{QA}) \cdot \mathbf{w} \\ &= \overrightarrow{PQ} \cdot \mathbf{u} + \overrightarrow{PQ} \cdot \mathbf{v} + \overrightarrow{PQ} \cdot \mathbf{w} = \overrightarrow{PQ} \cdot (\mathbf{u} + \mathbf{v} + \mathbf{w}) = 0, \end{aligned}$$

since $\mathbf{u} + \mathbf{v} + \mathbf{w}$ is the zero vector.

Hence

$$PD + PE + PF = QD + QE + QF.$$

Since P, Q are an arbitrary pair of points, it follows that $PD + PE + PF$ is a constant for all points P within the triangle. ■

Enter the cousin The clinching condition in the above proof is the fact that $\mathbf{u} + \mathbf{v} + \mathbf{w}$ is the zero vector. This observation allows us to find a new theorem — the ‘cousin’ of Viviani’s theorem.

The vector proof shows that if P is a point within an equilateral $\triangle ABC$, and $\mathbf{u}, \mathbf{v}, \mathbf{w}$ are *any* three fixed unit vectors such that $\mathbf{u} + \mathbf{v} + \mathbf{w}$ is the zero

vector, then the sum of lengths of the projections of PA , PB , PC on $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$ (respectively) will be a constant for all such points P .

But we get infinitely many theorems from this statement, because we can choose the three unit vectors $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$ in infinitely many ways! All we need to ensure is that their (vector) sum is zero.

Here is one possibility: Let $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$ be unit vectors along the directions $\overrightarrow{AB}$, $\overrightarrow{BC}$, $\overrightarrow{CA}$ respectively. It is obvious that the sum of these three unit vectors is zero.

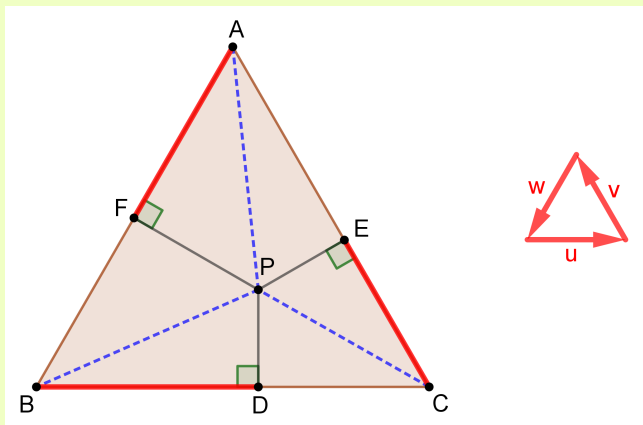


Figure 30

This gives rise to the following theorem (see Figure 30).

Theorem (Clough-de Villiers). *Let $\triangle ABC$ be equilateral, and let P be a point in its interior. Let perpendiculars PD , PE , PF be dropped to BC , CA , AB respectively. Then $BD + CE + AF$ has the same value for all positions of P .*

It is easy to deduce what the constant value of $BD + CE + AF$ must be.

Let P lie at the circumcentre of $\triangle ABC$; then $BD = CE = AF = a/2$ where a is the side of the triangle. This gives $BD + CE + AF = 3a/2$.

Hence an equivalent claim is:

$$BD + CE + AF = \frac{3a}{2} \quad \text{for all positions of } P.$$

This yields yet another form of the claim. For, if $BD + CE + AF = 3a/2$, then we also have $CD + BF + AE = 3a/2$. So:

$$BD + CE + AF = CD + BF + AE \quad \text{for all positions of } P.$$

Readers may want to see a proof of this theorem which is not vector based. We give one such proof below. The proof will be algebraic in nature.

Alternate proof of the claim

This proof is based on the Pythagorean theorem. We prove the claim in the form ' $BD + CE + AF = CD + BF + AE$ for all positions of P '.

Since $PB^2 = PD^2 + BD^2$, we have:

$$\begin{aligned} PB^2 - PC^2 &= BD^2 - CD^2 \\ &= (BD + CD) \cdot (BD - CD) \\ &= a(BD - CD). \end{aligned}$$

We have similar relations for $PC^2 - PA^2$ and $PA^2 - PB^2$. Hence:

$$\begin{cases} PB^2 - PC^2 = a(BD - CD), \\ PC^2 - PA^2 = a(CE - EA), \\ PA^2 - PB^2 = a(AF - FB). \end{cases}$$

The sum of the three quantities on the left is 0, and so therefore is the sum of the quantities on the right.

We therefore have,

$$(BD - CD) + (CE - EA) + (AF - FB) = 0,$$

and hence,

$$BD + CE + AF = CD + BF + AE.$$

It follows that $BD + CE + AF = 3a/2 = CD + BF + AE$. 

Further reading

- [1] Michael de Villiers, "An illustration of the explanatory and discovery functions of proof," *Pythagoras*, Vol 33, No 3, from <https://pythagoras.org.za/index.php/pythagoras/article/view/193/227>

Article contributed by Shailesh Shirali

Equilateral Triangles All the Way – a Diary Entry

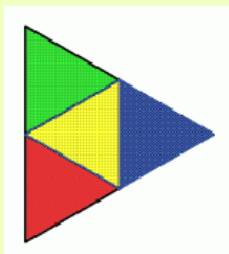
Article contributed by Priyadharshni Elangaivendan

This article describes an experience where students of class 8 in Sahyadri School, KFI (near Rajguru Nagar, Pune) were presented with a fascinating *low threshold high ceiling task* [1] from the NRICH website.

What is a low threshold high ceiling task? NRICH [2], [3] describes a “low threshold high ceiling task” to be a task “which is designed to be mathematically accessible, and to have built-in extension opportunities. In other words, everyone can get started and everyone can get stuck.” What is most appealing about such a task is that every child feels confident in taking up the challenge and coming up with some worthwhile observation or the other. We see this happening in the present case study.

A low threshold high ceiling task involving equilateral triangles

An equilateral triangle can be dissected into four (equal) smaller equilateral triangles.



- (a) Can you dissect an equilateral triangle into 6 smaller ones, not necessarily all the same size?
- (b) Can you dissect an equilateral triangle into 15 smaller ones, not necessarily all the same size?

Figure 31: A low threshold high ceiling task involving equilateral triangles

The students got involved in the challenge almost immediately, working individually and silently. Gradually, they started to share their ideas. Some asked if they could dissect the equilateral triangle into 16 smaller ones instead of 15, as they found an easy solution (Figure 2).

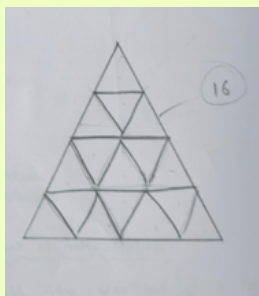
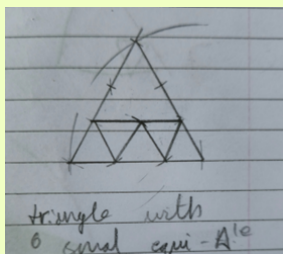


Figure 32: Dissection into 16 equilateral triangles

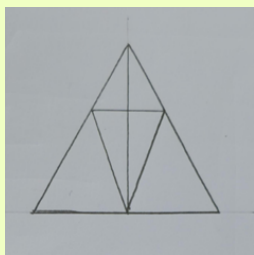
Considering the variety of possible designs that may not qualify as a solution for the given question, the students were instructed to record their drawings, even if it was not a solution for 6 or 15 (but was, say, for 19).



One of the first solutions for the number 6 included a construction using a compass.

Figure 33: Dissection into 6 equilateral triangles

As noted earlier, an attractive aspect of a low threshold high ceiling task is every child feels confident in taking up the challenge. For both 6 and 15, the first few solutions came from students who were usually silent in the class.



A few clarifications had to be made before everybody was on the same page. Though we had mentioned that the task required dissection into 6 equilateral triangles, some children presented solutions in which the smaller triangles were not equilateral.

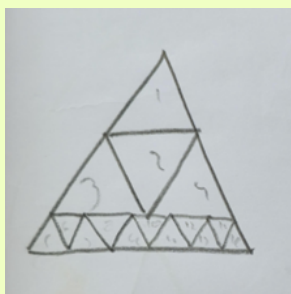
Figure 34: An equilateral triangle split into 6 triangles, but not all are equilateral

Figure 34 illustrates how the student built the solution for dissection into 6 equilateral triangles from Figure 31 (given in the question) by drawing a straight line through the middle. This provided an opportunity to discuss some geometric properties of right-angled triangles (not related to this problem).

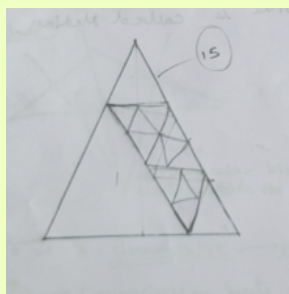
It had to be reiterated many times that the regions in the dissection had all to be only equilateral triangles.

One student counted the total number of equilateral triangles in the figure rather than the number of equilateral triangles in the dissection. Thus, one can count 5 equilateral triangle in Figure 31 if one includes the outer triangle. This shows how easily instructions can be misunderstood.

For half the session, students worked on specific cases like 6, 15 and other numbers of their choice. By the middle of the session they had obtained many designs, corresponding to different numbers, and also different designs for the same number (Figure 35).



(a)



(b)

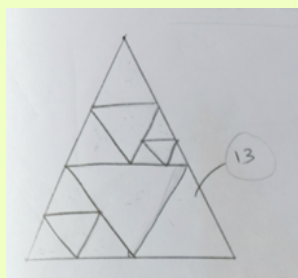
Figure 35: Two different designs for the number 15

Extending the exploration

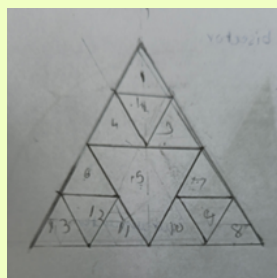
As the students generated more and more designs, they started announcing the numbers they had designs for, and saying things like “it just goes on forever; we can draw designs for any given number,” and so on. At this point we started recording the various numbers for which we had a solution. At the start, students were looking only for designs for specific numbers, but now some of them started making generalized observations. This became possible because the class now had a rich collection of designs. Some of these generalized observations are listed below.

Observation 1: *Given any number n , there may be multiple ways of dissecting the given triangle to get n equilateral triangles.*

E.g., see Figure 35 for two different designs for the number 15; or see Figure 36 for two different designs for the number 13.



(a)



(b)

Figure 36: Two different designs for the number 13

Observation 2: *Given any design, one can add 3 more triangles to the existing design by adding an inverted triangle inside any existing triangle.*

Figure 37 represents the drawing a student made while she was explaining how one can add 3 more triangles to the existing number. She started with 11 triangles, drew a triangle inside a triangle giving 14 equilateral triangles, to this drawing she added another triangle giving 17 equilateral triangles, and so on. This way, she argued, the process can go on infinitely.

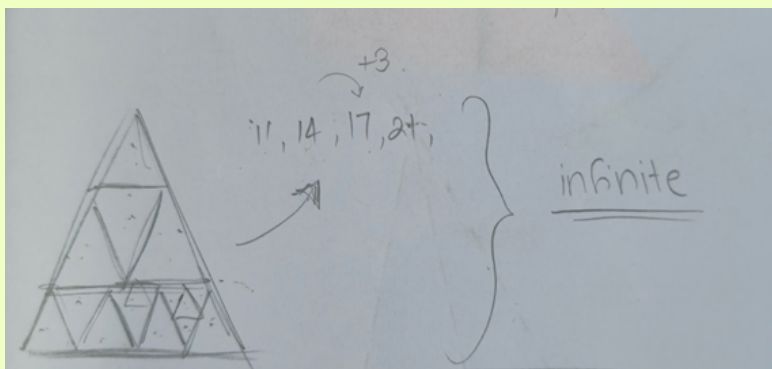


Figure 37: Adding 3 more triangles

Observation 3: *Let n be any even number greater than 2. Then an equilateral triangle can be dissected into n equilateral triangles.*

To see why, we write $n = 1 + (n - 1)$. Here, $n - 1$ is an odd number, and $n - 1 \geq 3$ since $n \geq 4$. We now fit $n - 1$ identical equilateral triangles to form a solid 'bottom row' of the given triangle. This naturally leaves a triangular space above the row; it too is equilateral. This design gives the desired dissection (see Figure 38; this is for the case $n = 4$).

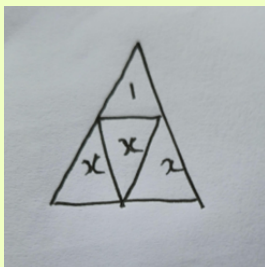
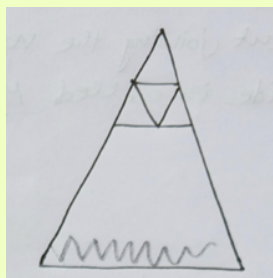


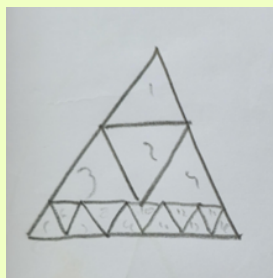
Figure 38: Student's representation of the above observation

Observation 4: *Let n be any odd number greater than 5. Then an equilateral triangle can be dissected into n equilateral triangles.*

To see why, write $n = 4 + (n - 4)$. Here, $n - 4$ is an odd number, $n - 4 \geq 3$. Fit $n - 1$ identical equilateral triangles to form a solid 'bottom row' of the given triangle. Next, dissect the triangular space above the row into 4 equilateral triangles. This yields the desired dissection (see Figure 39).



(a)



(b)

Figure 39: (a): Illustration of the idea. (b) A concrete case, $n = 15$

Observation 5: *Let n be any number other than 2, 3 or 5. Then an equilateral triangle can be dissected into n equilateral triangles.*

The students were able to arrive at this remarkable conclusion by studying their various designs and combining the data with the above observations.

Observation 6: *A given equilateral triangle can be dissected into infinitely many smaller equilateral triangles.*

This follows from observations 3 and 4 (above). Or study Figure 40; this shows how a small parallelogram in the bottom row can again be split into a few triangles and a parallelogram. The resultant parallelogram can again be split in a similar way; and we can continue this way for ever.

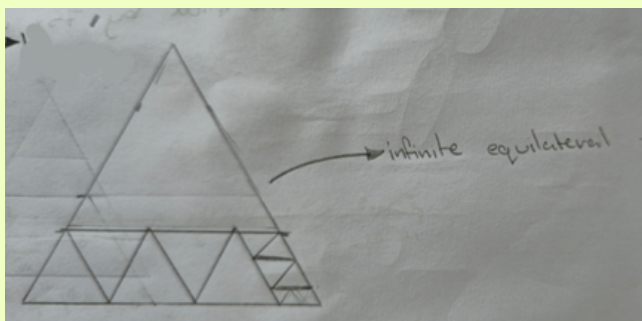


Figure 40: An equilateral triangle split into infinitely many equilateral triangles

Closing remarks

The lesson proved to be a happy collaborative exercise in which the collection of solutions and non-solutions enabled the children to build on

each others' observations and designs to come up with the generalizations noted above. The task motivated the students to think about triangles all the way inside!

Apart from being a happy classroom experience, the lesson also illustrates very nicely the potential of low threshold high ceiling tasks.

Further reading

[1] NRICH, "Cut it out" from

<https://nrich.maths.org/problems/cut-it-out>

[2] NRICH, "Enriching mathematics for all learners" from

<https://nrich.maths.org/>

[3] NRICH, "Low threshold high ceiling" from

<https://nrich.maths.org/low-threshold-high-ceiling>

About the author

Priyadharshni Elangaivendan is a mathematics teacher and education researcher, currently teaching at Sahyadri School, Pune. She is passionate about creating exploratory learning spaces for children. She is also pursuing a PhD in Mathematics Education at the Homi Bhabha Centre for Science Education, Mumbai. Her research interest lies in understanding the role of the body and environment in learning mathematics. She is currently designing touch- and sensor-based embodied technologies for learning mathematics and exploring how they contribute to conceptual understanding. She may be contacted at epriyadharshni@gmail.com.

How Do We Know That a Number Is Prime? – 1

In previous articles of this series, we looked at strings of prime numbers in arithmetic progression. Now we ask a more basic question: *How do we know whether a given number is prime or not?* This question quickly takes us into the study of algorithms and computational complexity.

Some numbers are ‘visibly composite.’ For example, if a number larger than 2 has units digit in the set $\{0,2,4,6,8\}$, it is composite. Similarly, if a number larger than 5 has units digit 5, it is composite. (*Assumption.* We are working throughout in base 10.) These tests correspond to divisibility by 2 and by 5, respectively. Checking for divisibility by 3, 9 and 11 involves a little more work but is still very easy. In this manner, we can quickly check whether a given number is divisible by any of the primes below 15. (You may not be familiar with the tests for divisibility by 7 and 13, but there are simple tests available which we discuss separately.)

But what happens next? If the given number clears all of these tests, do we declare the number to be prime?

Exercise You may want to try out this exercise with these four numbers:

$$5267, \quad 5273, \quad 5279, \quad 5287.$$

Two of these numbers are prime and the other two are composite. Which ones are the prime numbers? After spending some time on this exercise, you will begin to see that testing primality is a lot of work! (We will reveal the answers at the end of this article!)

Let's give the answer to the question posed above. If a number clears the tests of divisibility by 2,3,5,7,11,13, do we declare it to be prime? Certainly not!

It is quite easy to construct counterexamples to the claim that if a number is not divisible by 2,3,5,7,11,13, then it must be a prime number. Here is an example. We form the number

$$N = (2 \times 3 \times 5 \times 7 \times 11 \times 13) + 1 = 30031$$

and check whether it is prime. Well, it turns out to be composite!

$$30031 = 59 \times 509.$$

This factorization would be difficult to find on one's own.

The computations shown above point to the following conclusions.

- To prove that a given integer $N > 1$ is composite, all we need to do is exhibit a pair of integers $a > 1$ and $b > 1$ such that $ab = N$.
- To find such a pair of integers is, however, a very different matter! Indeed, this is computationally very demanding task.
- But how would one establish that a given integer M is prime? One could — in principle — try out all possible factorizations, and if none

of them work out, conclude that M is prime. But this is clearly a very demanding task, computationally. Imagine applying it to a number such as

$$M = 2^{31} - 1 = 2147483647,$$

which, by the way, *is* prime. Think of the number of computations involved and how long such a task would take. This approach would be wholly unfeasible and impractical. We need to find some better approach.

The approach that suggests itself at this stage is to find some property that is *uniquely characteristic* of the primes; in other words, a property that primes possess and composite numbers do not possess. We consider this question below.

Wilson's theorem

Here is a property of prime numbers that exactly fits the requirement stated above. It has been known for a long time. Wikipedia [1] tells us that it was first stated in around 1000 AD by the Arab mathematician Ibn al-Haytham, but that “it is named after John Wilson, an English mathematician, as the theorem was published by his teacher Edward Waring in 1770, who attributed it to Wilson.” The statement was later proved by Joseph-Louis Lagrange, in 1771. Here is the statement of the theorem:

Theorem (Wilson). *For any prime number p , we have,*

$$(p - 1)! \equiv -1 \pmod{p}.$$

In other words, $(p - 1)! + 1$ is a multiple of p .

Numerical illustration of the theorem

The table below illustrates the claim of the theorem. For convenience we use the symbol a_n to denote the remainder in the division $(n-1)! + 1 \div n$.

n	2	3	4	5	6	7	8	9	10	11	12	13	14	15
a_n	0	0	3	0	1	0	1	1	1	0	1	0	1	1

Computing these remainders requires some effort but not as much as you may think. For example, let us see how to compute a_{16} . We require the remainder in the division $(15! + 1) \div 16$. All computations shown below are carried out modulo 16.

$$3! = 6 \equiv 6, \quad 4! \equiv 6 \times 4 \equiv 8, \quad 5! \equiv 8 \times 5 \equiv 8,$$

$$6! \equiv 8 \times 6 \equiv 0, \quad 7! \equiv 0 \times 7 \equiv 0, \quad 8! \equiv 8 \times 0 \equiv 0,$$

etc., so $n! \equiv 0 \pmod{16}$ for $n \geq 6$. Hence $15! + 1 \equiv 1 \pmod{16}$.

Another example: let's compute a_{19} . All computations shown below are carried out modulo 19.

$$3! = 6 \equiv 6, \quad 4! \equiv 6 \times 4 \equiv 5, \quad 5! \equiv 5 \times 5 \equiv 6,$$

$$6! \equiv 6 \times 6 \equiv -2, \quad 7! \equiv (-2) \times 7 \equiv 5, \quad 8! \equiv 5 \times 8 \equiv 2,$$

$$9! \equiv 2 \times 9 \equiv -1.$$

Next, observe that modulo 19,

$$10 \equiv -9, \quad 11 \equiv -8, \quad 12 \equiv -7, \quad 13 \equiv -6, \quad \dots, \quad 18 \equiv -1,$$

implying that

$$\begin{aligned} 10 \times 11 \times 12 \times \cdots \times 18 &\equiv (-9) \times (-8) \times (-7) \times \cdots \times (-1) \\ &\equiv (-1)^9 \cdot 9! \equiv -1 \times -1 \equiv 1 \pmod{19}, \end{aligned}$$

hence

$$18! \equiv -1 \times 1 \equiv -1 \pmod{19}, \quad \therefore 18! + 1 \equiv 0 \pmod{19},$$

and so, $a_{19} = 0$.

That required a fair bit of work but we were spared the pain of having to compute $19!$ in full.

Exercise

1. Compute the following: a_{20}, a_{21}, a_{22} .
2. Verify that $a_{23} = 0$.
3. If you examine the table shown above, you will notice that a_n takes the value 1 very often. Why should this be so? Try to provide a justification.

Remark

Wilson's theorem is a very striking result. What is remarkable about it is that it uniquely characterizes the primes. But as a test for primality it is wholly impractical. The computational work involved is much too large. Just imagine the number of computations required to check the primality of a 10-digit number! And what will we do with still larger numbers? We must clearly look for computationally feasible alternatives.

What about the proof of Wilson's theorem?

You may wonder: how do we justify such a proposition? It seems most likely that the original discoverers of the theorem simply observed the pattern and then stated it as a proposition; the proof came later. You may be intrigued by the fact that as great a philosopher-mathematician as

Gottfried Leibniz was unable to find a proof (he empirically discovered the result). A few decades later, the French mathematician Louis Lagrange devised an elegant and ingenious proof. We shall elaborate on this proof in the next issue. We shall also study another such result which is even more striking; it was first discovered by Pierre de Fermat. That result turns out to have profound consequences that are at work even today.

Answer to the first exercise We had asked you to investigate the four numbers 5267, 5273, 5279, 5287. Well, these two are prime: 5273 and 5279; and other two are composite:

$$5267 = 23 \times 229, \quad 5287 = 17 \times 311.$$

Further reading

- [1] Wikipedia, “Wilson’s theorem” from
https://en.wikipedia.org/wiki/Wilson%27s_theorem
- [2] Weisstein, Eric W. “Wilson’s Theorem.” From MathWorld–A Wolfram Web Resource.
<https://mathworld.wolfram.com/WilsonsTheorem.html>
- [3] Wikipedia, “List of prime numbers” from
https://en.wikipedia.org/wiki/List_of_prime_numbers
- [4] Chris Caldwell, “The PrimePages: prime number research & records”
from <https://t5k.org/>

Article contributed by the editors

Teaching Algebraic Expressions Using the Area Model – 2

Article contributed by Tanuj Shah

In the last issue we saw how pattern language can be used as an entry point for teaching expressions.

Another entry point could be the *area model* using the area of a rectangle as our starting point.

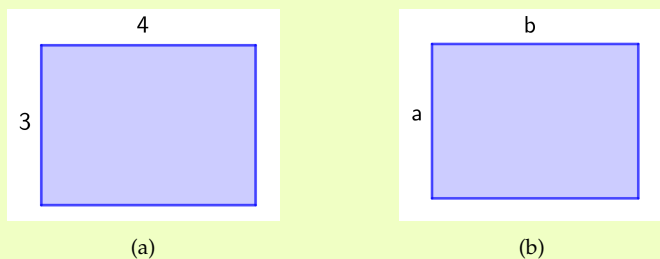


Figure 41

The area of the rectangle (Figure 41) can be written as 3×4 as the width and length are 3 and 4, while for the second rectangle it can be written as $a \times b$ or ab for short.

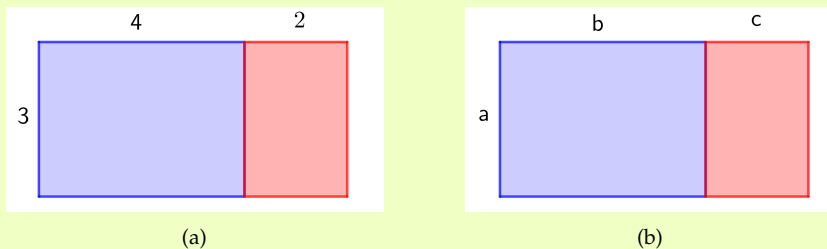


Figure 42

It can be followed by introducing the distributive law.

The area of the first rectangle (Figure 42) can be expressed as $3 \times 4 + 3 \times 2$.

The area of the second rectangle can be expressed as $a \times b + a \times c$ or $ab + ac$ for short.

This could be followed by showing the binomial identity for $(a + b)^2$.

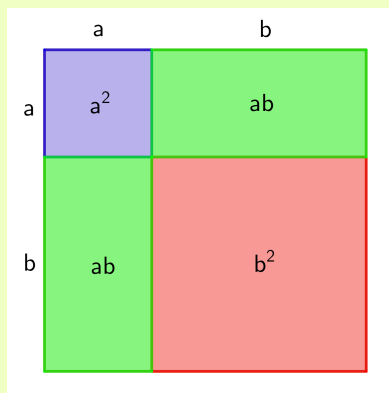
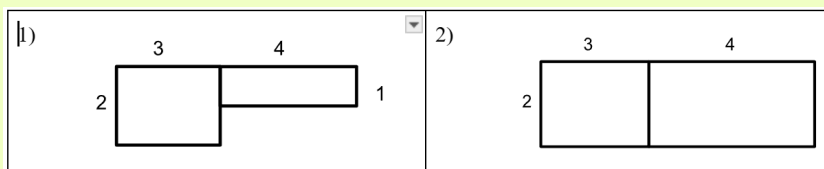


Figure 43

Exercises on area diagrams with numeric expressions

The students could work together in groups for these exercises. Draw each area diagram in your notebook, leaving some space between each diagram. Match each expression from the list below and write it underneath each diagram. There are two expressions for each diagram. There will be two expressions left in the end: draw the area diagram for these and write the two expressions under it. (All measurements are in units)



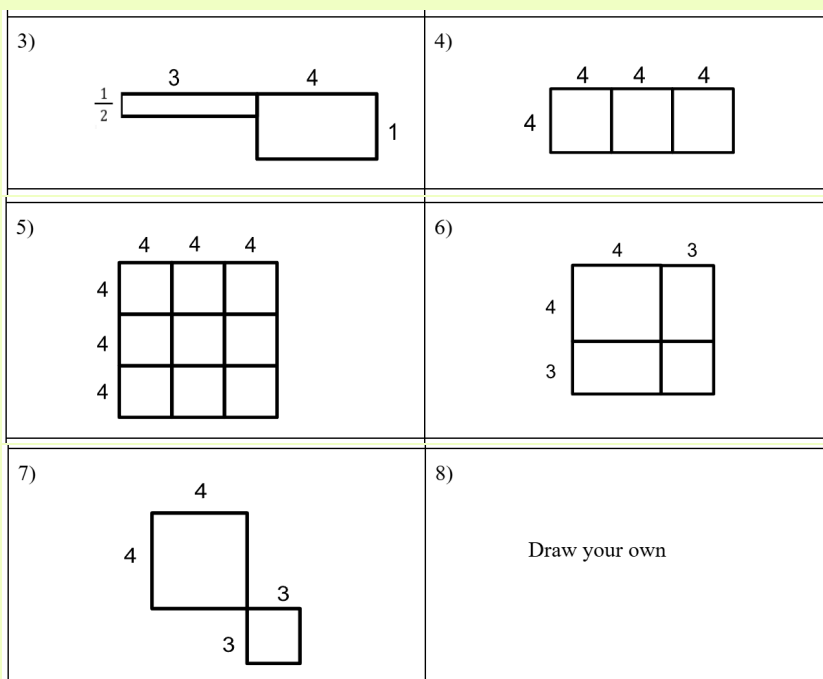


Figure 44

Expressions

1. $3^2 + 4^2$

6. $\frac{3}{2} + \frac{4}{2}$

2. $(2 \times 3) + (2 \times 4)$

7. $2 \times (3 + 4)$

4. 3×4^2

8. $4 + (3 \times 2)$

5. $(3 \times 4)^2$

9. $(3 + 4)^2$

10. $4 + \left(3 \times \frac{1}{2}\right)$

11. $\frac{1}{2} \times (3 + 4)$

12. $3^2 + 4^2 + (2 \times 3 \times 4)$

13. $\frac{3}{2} + 4$

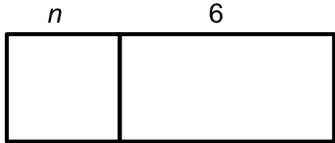
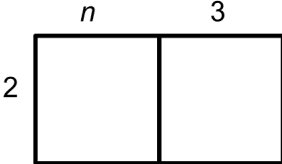
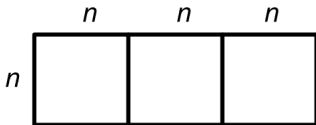
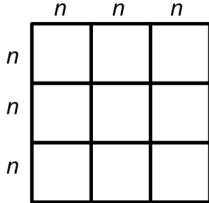
14. $(4 \times 3) + (4 \times 3)$

15. $(2 \times 3) + (1 \times 4)$

16. $(4 + 3)^2 - 2 \times (4 \times 3)$

Exercises on area diagrams with algebraic expressions

(A) Draw each area diagram in your notebook, leaving space between the diagrams. (All measurements are in units)

1) 	2) 
3) 	4) 

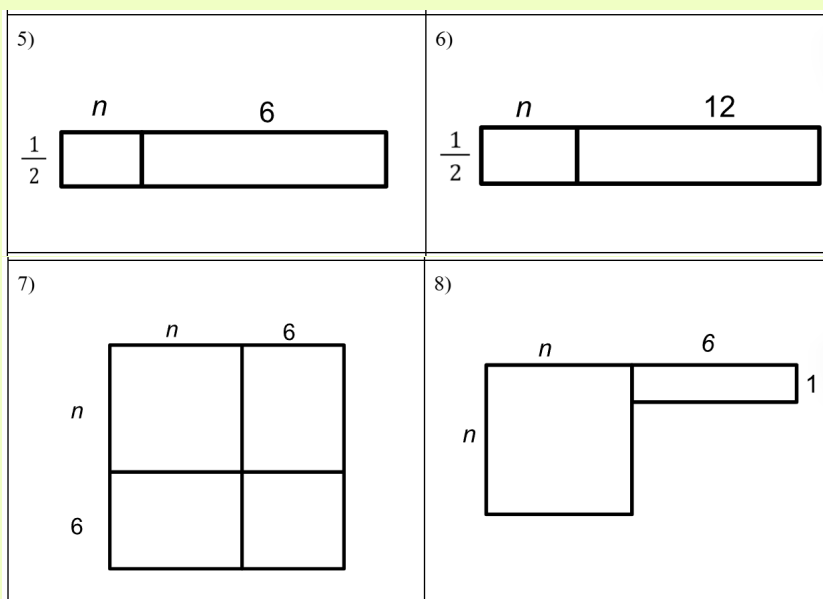


Figure 45

(B) Match the following expressions to the area diagrams above:

1. $\frac{n+6}{2}$

5. $2(n+3)$

9. $n^2 + 12n + 36$

2. $3n^2$

6. $\frac{n}{2} + 6$

10. $\frac{n}{2} + 3$

3. $2n + 12$

7. $(3n)^2$

11. $n^2 + 6$

4. $2n + 6$

8. $(n+6)^2$

(C) Match the following sentences to the diagrams above:

1. Multiply n by two, then add six.

2. Multiply n by three, then square the answer.
3. Add six to n , then multiply by two.
4. Add six to n , then divide by two.
5. Add three to n , then multiply by two.
6. Add six to n , then square the answer.
7. Multiply n by two, then add twelve.
8. Divide n by two, then add six.
9. Square n , then add six.
10. Square n , then multiply by 3.
11. Square n , then multiply by 9.

(D) Taking the value of n as 4, work out the area for each diagram.

About the author

Tanuj Shah did his teacher training at Nottingham University and taught in various schools in England before joining Rishi Valley School. He taught Mathematics in Rishi Valley School for over two decades, where he developed hands-on self learning-modules for the Junior and middle School. He has a deep passion for making mathematics accessible and interesting for all. He may be contacted at shahtanuj@hotmail.com

A curious property of the multiples of 999

Article contributed by Hans Humenberger

While investigating a curiosity of the number 2997 (see [\[1\]](#), [\[2\]](#)), we needed to find the digit sums of the multiples of 999 (the numbers 999, 1998, 2997, 3996, ...).

We observed that the first several multiples all have digit sum $3 \times 9 = 27$, and we needed to verify how far this constancy holds. We did not think about the problem further but simply used a computer, expecting that after some multiples, the digit sum would change. But there was no change within the first 1000 multiples; all of them have digit sum 27, an unexpected curiosity!

Of course, the constancy must break down after this, since

$$999 \times 1001 = 999999,$$

with digit sum $6 \times 9 = 54$.

Having discovered this fact empirically, we asked, why? This question cannot be answered by computer experiments. We need to achieve insight into the structure of the problem.

As we will see, this is not difficult. Indeed we are able to prove the result in two different ways.

Version 1

If $k = 1000$ there is nothing to prove since $999 \times 1000 = 999000$, with digit sum 27. So, let k be a number with three or fewer digits a, b, c , so $k = 100a + 10b + c$; here $0 \leq a, b, c \leq 9$ (not all are 0).

We need the sum of the digits of $999k$. Write $999k = 1000k - k$. The subtraction is carried out as shown below. We have assumed here that $c > 0$.

Place value $\rightarrow$	10^5	10^4	10^3	10^2	10	1
Digits of $1000k$	a	b	c	0	0	0
Digits of k				a	b	c
Digits of $999k$	a	b	$c - 1$	$9 - a$	$9 - b$	$10 - c$

We see that if $c > 0$, the digit sum of $999k$ is

$$a + b + (c - 1) + (9 - a) + (9 - b) + (10 - c) = 9 + 9 + 9 = 27,$$

as required.

If $c = 0$ and $b > 0$, the table looks like this:

Place value $\rightarrow$	10^5	10^4	10^3	10^2	10	1
Digits of $1000k$	a	b	0	0	0	0
Digits of k				a	b	0
Digits of $999k$	a	$b - 1$	9	$9 - a$	$10 - b$	0

We see that if $c = 0$ and $b > 0$, the digit sum of $999k$ is

$$a + (b - 1) + 9 + (9 - a) + (10 - b) + 0 = 9 + 9 + 9 = 27.$$

The reader should verify that the same outcome results if $c = 0, b = 0, a > 0$. (In this case the digits of $999k$ are, from left to right, $a - 1, 9, 9, 10 - a, 0, 0$.)

Having covered all possible cases, we are now sure of the result and we also have an explanation behind it. ■

Version 2

This proof does not need consideration of different cases. But we do require that $k > 0$ (equivalently, a, b, c are not all 0).

Place value $\rightarrow$	10^5	10^4	10^3	10^2	10	1
$1000k$	a	b	c	0	0	0
k				a	b	c
$999k$	$k - 1$			$1000 - k$		

As earlier, we write $999k = 1000k - k$. We then 'slice' the product into two blocks of three digits each as shown in the above array. Then the block

on the right represents the number $1000 - k$, and the block on the left represents the number $k - 1$. (This is so because of the exchanges that have happened from higher place-value columns to lower place-value columns while carrying out the subtraction.)

Now observe that the sum of these two three-digit numbers is

$$(k - 1) + (1000 - k) = 999.$$

An illustration with numbers will make this clearer. E.g., let $k = 472$. Then $1000k - k = 472000 - 472 = 471528$. Observe that $528 = 1000 - 472$, and $471 = 472 - 1$. And, of course, $471 + 528 = 999$.

The reader could try this out with some other number, e.g., $k = 360$, and verify the claim.

Now consider the addition of a pair of three-digit numbers whose sum is 999. Let's suppose it looks like this:

$k - 1$	d	e	f
$1000 - k$	g	h	i
Sum	9	9	9

In the rightmost column we must have $f + i = 9$, with no carry over (one cannot add two digits to get 19; the maximum possible value would be 18), so we start without carry over in the middle column and obtain, like earlier, $e + h = 9$, and finally $d + g = 9$ as well; the same logic works in each case. Therefore the sum of the digits in $999k$ is equal to

$$\begin{aligned} & (d + e + f) + (g + h + i) \\ &= (d + g) + (e + h) + (f + i) = 9 + 9 + 9 = 27. \end{aligned}$$

Generalization

We will naturally want to examine whether a similar property holds for the multiples of 99, 9999, 99999, and so on. Students could carry out this exploration on their own. They will soon discover that

- The numbers $99k$ all have digit sum $2 \times 9 = 18$ for $k = 1, 2, \dots, 100$;
- The numbers $9999k$ all have digit sum $4 \times 9 = 36$ for $k = 1, 2, \dots, 10000$;

and so on. The generalized result may be proved the same way as above, by writing $99k = 100k - k$, etc.; both versions of the proofs carry through.

Version 1

The display below shows the relevant details for the case when k is a four-digit number (i.e., $n = 4$) and the units digit d is non-zero.

Place value $\rightarrow$	10^7	10^6	10^5	10^4	10^3	10^2	10	1
Digits of $10000k$	a	b	c	d	0	0	0	0
Digits of k					a	b	c	d
Digits of $9999k$	a	b	c	$d - 1$	$9 - a$	$9 - b$	$9 - c$	$10 - d$

We see that if $d > 0$, the digit sum of $9999k$ is

$$\begin{aligned}
 & a + b + c + (d - 1) + (9 - a) + (9 - b) + (9 - c) + (10 - d) \\
 &= 9 + 9 + 9 + 9 = 36,
 \end{aligned}$$

as required.

Similar displays can be shown for the cases when $d = 0$, $c > 0$, or $c = d = 0$, $b > 0$, or $b = c = d = 0$, $a > 0$.

Version 2

We also give the display for to the second version of the proof in which a study of separate cases is not required.

Place value →	10^7	10^6	10^5	10^4	10^3	10^2	10	1
10000 <i>k</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	0	0	0	0
<i>k</i>					<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>
9999 <i>k</i>	<i>k</i> – 1				10000 – <i>k</i>			

We write $9999k = 10000k - k$ and ‘slice’ the product into two blocks of four digits each (see the above array). The block on the right represents the number $10000 - k$, and the block on the left represents the number $k - 1$. The sum of the two numbers is 9999. Arguing the same way as we did earlier, we conclude that the sum of the four digits on the left and the four digits on the right is $4 \times 9 = 36$.

Closing remark

Having observed this phenomenon empirically, we have been able to find a convincing explanation and proof using the mechanism of the subtraction algorithm in the place value system. It is a vivid demonstration of the power of mathematics.

Before seeing the phenomenon, very few people would have expected that the first one thousand multiples of 999 all have the same digit sum 27. Surprising things are always motivating!

Further reading

- [1] Humenberger, H. *Kohli's Number 2997 and corresponding proofs*. To appear in *The Compass*.
- [2] Kohli, U. (2020). *The Mystical Number 2997 (Kohli's Number)*. *At Right Angles* (8). pp. 35-36. <https://publications.azimpremjiuniversity.edu.in/2588/>

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The Enduring Benefits of Handwriting for Memory and Learning

Adapted from: <https://www.scientificamerican.com/article/why-writing-by-hand-is-better-for-memory-and-learning/>. Author: Charlotte Hu (edited by Lauren J. Young)

Despite the increasing integration of digital technology in education, research consistently highlights the superior benefits of handwriting for memory and learning, particularly for young children.

A recent study on brain activity during note-taking revealed significantly higher electrical activity across a wider network of brain regions in students writing by hand compared to those typing. This network has areas responsible for movement, vision, sensory processing, and memory, underscoring the profound cognitive engagement involved in handwriting.

This new research builds upon earlier findings suggesting that typing often leads to a less thoughtful transcription of information, bypassing deeper cognitive processing.

Conversely, handwriting necessitates active engagement with the material. Students must pay attention, process information, prioritize key concepts, and connect them to prior knowledge due to the physical limitations of writing speed. This conscious effort to synthesize information enhances comprehension and retention.

The NTNU researchers utilized EEG technology to compare brain activity as students wrote or typed words. Handwriting triggered widespread connectivity across visual, sensory, and motor cortices. The motor cortex, responsible for body movement and sensorimotor integration, is significantly more activated during handwriting. Professor Audrey van der Meer explains that the unique motor sequence for each handwritten letter provides a distinct bodily feel, unlike the repetitive finger movements in typing. She notes that children who primarily use digital tablets for writing may struggle with differentiating visually similar letters due to this lack of varied motor engagement.

Educational neuroscientist Sophia Vinci-Booher corroborates these findings, emphasizing the strong coupling of motor and sensory systems in handwriting. The act of forming a letter involves a perceptual understanding translated into a motor action, which is then visually processed again, strengthening the association between the action and the concept. This tangible creation reinforces memory, similar to how physically realizing an imagined idea enhances its retention.

The memory-enhancing effect of physical creation is well-documented. Studies show that writing, drawing, or enacting words prompts deeper

engagement with the information. Cognitive neuroscientist Yadurshana Sivashankar explains that handwriting requires more intricate activation of the brain's motor programs compared to typing, where the hand movements are less directly related to the word's structure. Research demonstrates that performing actions related to learned information improves memory recall, highlighting the benefit of transforming information into a meaningful output that strengthens neural connections.

The implications for early childhood education are significant. Studies indicate that children learn letters and visual concepts more effectively through the coordinated hand movements of writing and drawing, an engagement absent in clicking or tapping. Vinci-Booher's research shows that handwriting engages different brain regions more profoundly than passive learning and leads to more lasting learning effects, particularly in letter recognition. Furthermore, the motor engagement in handwriting may be crucial for overcoming "mirror invariance," helping children distinguish between letters like 'b' and 'p'.

While acknowledging the utility of technology for certain educational tasks and equitable access, experts caution against over-reliance. "Cognitive offloading," the increasing dependence on digital devices for memory and navigation, reduces cognitive effort and may lead to a decline in memory and motor skills. Van der Meer expresses concern about the move towards fully digital schools, noting that young children are entering school with underdeveloped fine motor skills due to a lack of hands-on activities.

In conclusion, the evidence strongly supports the crucial role of handwriting in fostering memory and learning, especially in early childhood. The intricate motor and cognitive engagement during handwriting creates stronger neural connections and enhances information retention more ef-

fectively than typing. While technology has its place, the unique cognitive benefits of handwriting warrant its continued importance in education, particularly in developing foundational literacy skills.

Why Is the Night Sky Dark?

You Can Thank the Big Bang

From: <https://www.scientificamerican.com/article/why-is-the-night-sky-dark-olbers/>. Author: Phil Plait

The seemingly simple mystery of the sky's darkness is something that stumped astronomers for centuries

Quoting from Phil Plait's beautifully written article:

"I love it when a seemingly simple question—one so simple that it feels silly to even ask it—leads to profound, even cosmic, insight.

For example: Why is the sky dark at night?

I can imagine you reading this and thinking, "Seriously? *That's* profound?"

Yes. Yes, it is. I can also imagine you thinking, "But the answer's

obvious! There are only so many stars to go around. Ones farther away are fainter. So there's lot of black between them and nothing out there that they would illuminate. Of course the sky is dark at night."

But that answer is wrong or, at best, incomplete, and the actual situation wasn't obvious even to eminent scientists a century or so ago. And the person who first solved this mystery—or, at least, the first to give a correct scientific answer in writing—is almost certainly not who you'd think!

To be clear, the sky isn't *perfectly* dark at night. Earth's atmosphere glows faintly even from the darkest site. Here we're talking cosmologically dark, however: the universe itself is in a state of being unlit. There are some sources of background light even then—distant stars and galaxies too small and faint to be seen individually—yet the sky looks pretty black compared with, say, the surface of the sun.

That may seem like an extreme comparison, but it's not. That's the heart of the problem."

The author (Phil Plait) goes on to say that the historical assumption in astronomy was that the universe was infinite in both time and space, a static and unchanging entity that had always existed and always would. This view, particularly prevalent in the 19th century when the Milky Way was considered the entirety of the cosmos, seemed logical given the seemingly constant presence of stars in the night sky, contrasting with the more dynamic behaviour of the moon and planets.

However, this concept of an infinite and uniformly populated universe

leads to a significant contradiction known as Olbers' paradox. If the universe extends infinitely in space with stars distributed evenly throughout, then the night sky should be overwhelmingly bright. Consider concentric spherical shells centered on Earth. While a shell twice as distant would contain four times the number of stars due to its larger volume (increasing with the square of the distance), the brightness of individual stars within that shell would decrease by a factor of four due to the inverse square law of light. This cancellation implies that every shell, regardless of its distance, would contribute the same amount of brightness to our sky.



Figure 46: Source: Getty Images <https://www.scientificamerican.com/article/why-is-the-night-sky-dark-olbers/>

Extrapolating this to an infinite universe with an infinite number of such shells and stars, any line of sight from Earth, no matter how narrow, should eventually intersect the surface of a star. Consequently, the entire sky should appear as luminous as the surface of a star, a stark contrast to the darkness we observe at night.

One might propose that interstellar nebulas, composed of dust and gas, could obscure the light from distant stars, thus explaining the dark sky. However, this explanation falls short. These clouds would absorb the incoming radiation and eventually heat up, radiating energy themselves and becoming as bright as stars. Therefore, the introduction of nebulas merely delays the inevitable bright sky in an infinite and eternal universe.

This perplexing contradiction was formalized as **Olbers' paradox** by the German astronomer Heinrich Wilhelm Matthias Olbers in his 1823 treatise. However, the article points out that Olbers was not the first to ponder this issue; the question had been circulating for centuries, with the 16th-century English astronomer Thomas Digges being among the early thinkers who considered the implications of an infinite universe.

For a considerable period, no satisfactory scientific solution to Olbers' paradox emerged. The breakthrough came in 1848 with the publication of an essay titled "Eureka: A Prose Poem" by the American writer Edgar Allan Poe. While not providing a quantitative mathematical solution (which would later be developed by Lord Kelvin), Poe correctly proposed two key elements of the answer. Firstly, he suggested that the universe might not be infinite in extent. Secondly, acknowledging the finite speed of light, he reasoned that light from extremely distant objects simply had not had enough time to reach us since the universe's inception. The vast "voids which our telescopes find in innumerable directions" could be explained

by the immense distances of the invisible background, from which light has yet to arrive.

The second crucial part of the solution emerged in the early 20th century with advancements in astronomy. The realization that the universe was far larger than just the Milky Way, coupled with the groundbreaking discovery of the cosmic expansion, effectively discarded the static universe hypothesis. These observations paved the way for the Big Bang model. While the Big Bang model doesn't necessarily preclude an infinite universe in space, it does posit a beginning, a finite point in time from which the universe has been expanding. Consequently, we can only observe objects within a certain cosmic horizon, as light from objects beyond that distance has not yet had the time to travel to us.

Interestingly, the article notes that the expansion of the universe itself would contribute to the darkness of the night sky even in a hypothetical steady-state universe. As the universe expands, distant stars would recede at speeds that would cause their light to be redshifted beyond the visible spectrum or become too faint to detect. Therefore, while the dark night sky doesn't definitively prove the Big Bang, the Big Bang model provides a crucial framework for understanding this phenomenon.

Modern cosmology explains the dark night sky through a combination of factors: the finite age of the universe (limiting the distance from which light can reach us), the expansion of the universe (causing redshift and dimming of distant light), and the finite lifespan of stars (eventually leading to a universe devoid of luminous objects). As Lord Kelvin astutely observed, scientific paradoxes often arise from our limited understanding. As our knowledge of the universe deepens, these apparent contradictions dissolve, revealing a more coherent and comprehensible cosmic reality.

A passion for problem-solving

Adapted from: <https://twas.org/article/passion-problem-solving>. Author: Cristina Serra

Indian mathematician Neena Gupta received the 2023 TWAS-CAS Young Scientist Award for Frontier Science for providing a long-sought solution to an algebraic problem

Indian mathematician Neena Gupta has been awarded the prestigious 2023 TWAS-CAS Young Scientist Award for her groundbreaking solution to the long-standing Zariski Cancellation Problem (ZCP) in affine algebraic geometry. This award, established in 2020 and sponsored by Lenovo, recognizes the scientific achievements of young scientists under 45 working in developing countries. Gupta, who dreamed of a mathematics career since childhood, received the \$10,000 prize for her resolution of a problem first posed in 1949 by Oscar Zariski, a key figure in modern algebraic geometry.

The ZCP was considered a major challenge in the field, baffling mathematicians for decades. Gupta's success in 2014 marked a significant

breakthrough. She explained the problem in simplified terms as determining whether two original geometric objects have the same structure if their “cylinders” (objects with an added dimension) possess the same structure.

Gupta’s fascination with the ZCP began during her PhD studies at the Indian Statistical Institute (ISI) in 2011. Despite initial scepticism from a visiting mathematician who deemed it too difficult, she persevered. After three years of dedicated work, she achieved a complete solution. Mentors at the Tata Institute of Fundamental Research and ISI verified her work, leading to its publication. However, Gupta noted that many experts initially found it hard to believe that a young, relatively unknown scientist had solved such a significant problem.



Figure 47: Prof. Gupta lecturing at 89th Anniversary General Meeting of the Indian National Science Academy, on 6th December 2023. Source: <https://twas.org/article/passion-problem-solving>

While the ZCP may not have direct everyday applications, Gupta emphasizes its pivotal importance in advancing certain branches of knowledge with potential real-world impact, describing it as “a brilliant exercise of the mind with potential indirect real-life applications.”

Gupta views the TWAS-CAS award as a major life achievement, motivating her to further pursue fundamental mathematical problems. She believes such recognitions play a crucial role in shaping research priorities, revitalizing research themes, and increasing the visibility of work, thereby encouraging further contributions in the field.

The TWAS-CAS award is the latest in a series of accolades for Gupta’s work on the ZCP and her contributions to affine algebraic geometry and commutative algebra. Previous honours include the *Nari Shakti Puraskar* (India’s highest honour for women, 2022), the *Ramanujan Prize* (2021), and the *Shanti Swarup Bhatnagar Prize* (2019).

Currently a professor at ISI Kolkata, Gupta continues to develop new algebraic theories and explore unresolved mathematical problems. She actively encourages women in mathematics, emphasizing that the field is not exclusive to men and that perseverance is key to success, referencing the influential mathematician Emmy Noether as an example. While acknowledging the pressure that awards can bring, Gupta remains thrilled by the challenge of unsolved mathematical problems.

Can Earth's Rotation Be a Power Source?

Adapted from: <https://www.scientificamerican.com/article/can-earths-rotation-be-a-power-source-physicists-debate-bold-new-idea/>. Author: Elizabeth Gibney

A team of physicists has controversially proposed a novel method for generating electricity by harnessing the energy of Earth's rotation through its magnetic field. Led by Christopher Chyba of Princeton University, their findings, published in *Physical Review Research*, suggest that a specially designed device can produce a tiny voltage (17 microvolts) by exploiting the interaction between a conductor and Earth's static magnetic field as the planet rotates.

The concept, though intriguing, has been met with scepticism due to the minute voltage produced, making it challenging to rule out other potential causes. Emeritus physicist Paul Thomas finds the carefully conducted ex-

periments convincing, while retired physicist Rinke Wijngaarden remains unconvinced, having failed to replicate the effect in his own experiments.

The theoretical basis of the device lies in the principle of electromagnetic induction, similar to conventional power stations. As Earth rotates, a conductor on its surface moves through the planet's magnetic field. Normally, in a uniform field like Earth's, electrons would rearrange to create an opposing electric force, preventing a continuous current. However, Chyba and his colleagues theorized that a cylindrical tube made of specific materials with unusual magnetic properties could channel Earth's magnetic field into a unique configuration. This configuration, they argue, would generate a magnetic push that the internal electrostatic forces could not counteract, resulting in a current.

To test this, they constructed a hollow cylinder from a soft magnetic material. Their measurements revealed the predicted small voltage, dependent on the device's orientation relative to Earth's magnetic field, which was absent when a solid conductor was used.

Sceptics like Wijngaarden acknowledge the researchers' efforts to eliminate spurious effects but emphasize the need for further verification. He suggests testing the device at different latitudes to see if the results align with theoretical predictions.

Chyba acknowledges the need for independent verification and states that scaling up the device to produce usable energy would be the next step if the phenomenon is confirmed. Their calculations suggest that even meeting global electricity demands with this method would only negligibly slow Earth's rotation, comparable to natural effects. While the concept is provocative, more robust evidence is required to validate this potential source of emission-free power for remote applications or medical devices.

Why are prestigious universities falling prey to “Math crisis”?

Adapted from: <https://timesofindia.indiatimes.com/education/news/why-are-americas-prestigious-universities-falling-prey-to-math-crisis/articleshow/119862829.cms>. Author: Trisha Tewari

The original title of Trisha Tewari’s Times of India article was *Why are America’s prestigious universities falling prey to “Math crisis”?* But the article may be just as relevant for the Indian context, so we have adapted it suitably for the magazine. We request readers to judge for themselves.

America’s elite universities are facing a significant “math crisis” characterized by incoming students lacking basic mathematical skills. This issue,

highlighted recently by Harvard's introduction of remedial math courses, signals a systemic failure in K-12 education. National data from a 2024 report reveals a decline in math proficiency among eighth graders, with only 33% proficient, a drop from 41% in 2017. This decline disproportionately affects minority and low-income students. International PISA rankings also show a downward trend for the US.

Several factors contribute to this crisis. A shift in math education towards abstract concepts over practical application has left students weak in basic calculations. Over-reliance on technology has hindered the development of manual arithmetic skills. Inconsistencies in K-12 math education, coupled with promotion without mastery, have created foundational gaps. The COVID-19 pandemic and remote learning exacerbated these issues. Furthermore, the focus on standardized test scores often prioritizes test-taking strategies over genuine understanding. Finally, a disconnect between high school and college math curricula leaves many students unprepared for university-level rigour. This crisis threatens the future of education, innovation, and critical thinking, necessitating urgent systemic reform.

Points of relevance for the Indian context

We list below some questions based on the above report that we feel are of relevance for the Indian context. We request readers to examine them very carefully.

1. Is it true that our universities are experiencing a decline in the basic math skills of incoming students and are meeting this challenge by introducing remedial math courses? If so, what has been their experience?

2. Is our student population uniformly affected by this problem or are there some subgroups within our demography that have been affected to a greater extent than other groups?
3. Are the following true for the Indian context?
 - (a) “A shift towards conceptual learning over calculation ...” This may be true of the American context. Is precisely the opposite to draw in the Indian context?
 - (b) Over-reliance on technology.
 - (c) Pandemic learning loss; the carry-over from COVID.
 - (d) An exaggerated focus on examination marks over conceptual understanding and mastery of the material.
 - (e) A sharp discontinuity between high school mathematics and college mathematics.

We obviously need to give careful thought to this problem and act appropriately.

The Lotus Problem

Article contributed by Shashidhar Jagadeeshan

Introducing the problem

There are four temples belonging to four goddesses. The four temples are at the four corners of a temple tank.

A devotee wants to offer lotuses to the goddesses. They are quite jealous of each other so she has to offer the same number to each of them! Before she offers them, she needs to wash them in the temple tank. The tank is magical; every time she dips them in the water, they double in number. She also has to offer the goddesses flowers one at a time – she can't offer all four in one go.

So, to repeat, before offering any lotuses to a goddess she needs to dip the lotus in the tank, and every time she dips lotuses they double in number!

Moreover, she needs to offer each goddess the same number of lotuses and has to do this serially and not in parallel.

So, the question is: *How many lotuses should she start with, and how many should she offer to each temple?*

I was introduced to this rather amusing problem, which I will refer to as '**The Lotus Problem**', by my colleague who found it in a Hindi text she was using in class. I am unable to source where this problem originated, though it definitely has the flavour of a problem from the *Lilavati* by Bhaskaracharya! But it is more likely a problem from folklore whose origins are forgotten.

The problem is simple enough to state that it can be introduced to students as young as ten years old and they can solve it by trial and error. Even though trial and error seems rather elementary, students often find very interesting strategies to minimize their guesses, and discussing student strategies often reveals many insights. One can also tweak the problem by changing the number of temples and the number of times the flowers multiply after dipping.

Of course the more sophisticated approach is to introduce algebraic notation. Once we do this, some interesting mathematics emerges, very accessible to school students. The solution involves linear Diophantine equations and geometric series! The use of geometric series allows us to generalize the problem to having n temples instead of 4 and increasing the flowers m - fold, instead of doubling. The problem comes in handy as a nice application while teaching these topics at higher grades. One student claimed she will never forget the formula for the sum of the geometric series after working on this problem!

Pleasing just four goddesses!

We begin with the problem as stated in the introduction, that is $n = 4$ and $m = 2$.

Let x be the number of lotus flowers we start with. Let y be the number we offer each temple.

After offering at the first temple we have

$$2x - y$$

flowers left.

After offering at the second temple we have

$$2(2x - y) - y = 4x - 3y$$

flowers left.

After offering at the third temple we have

$$2(4x - 3y) - y = 8x - 7y$$

flowers left.

Finally after the fourth temple we have

$$2(8x - 7y) - y = 16x - 15y$$

flowers left. But we are supposed to have no flowers at the end of the fourth offering, hence,

$$16x - 15y = 0.$$

This is an *indeterminate* or *Diophantine equation* of the first degree, i.e., an equation of the form

$$ax + by = c, \text{ where } a = 16, b = -15 \text{ and } c = 0.$$

Remark Diophantus (250 CE) was a great Greek algebraist who worked on linear and higher order equations. In all probability, mathematicians of yore from different cultures worked on these equations, but the name “Diophantine equation” has stuck!

Such equations have an infinite number of solutions! Since we are talking about numbers of lotuses, we want both x and y to be positive integers, these of course can be easily found if we rewrite

$$y = \frac{16}{15}x.$$

Clearly, if we let $x = 15$, then we get $y = 16$ and we get the smallest numbers of flowers we should start with and the number of flowers we should offer. The reader can easily check that if we started with 15 flowers and offered 16 each time, then we solve the puzzle. If we let $x = 15k$, where k is a positive integer, then $y = 16k$. And we find all the solutions we want with x and y being positive integers.

Is this the end of the story? Well, not if we want to generalize the problem. Also, the above method does not immediately tell us the connection between n , m and the solution. That is, in this case, what have 16 and 15, got to do with $n = 4$ and $m = 2$? Some readers might have figured out that $16 = 2^4$ and maybe that in this case $y - x = 1$. But is this always the case? Let us see what happens when $n = 4$ and $m = 3$, that is we have four temples, but the flowers triple each time they dip.

Tripling the problem

As before we let x be the number of lotus flowers we start with and let y be the number we offer each temple. This time however $n = 4$ and $m = 3$.

After offering at the first temple we have

$$3x - y$$

flowers left.

After offering at the second temple we have

$$3(3x - y) - y = 3^2x - y(1 + 3)$$

flowers left.

You might have noticed that we have expressed the coefficients of x and y as powers of 3 and sums of powers of 3 (here, $1 = 3^0$). This is on purpose, and we will soon see why.

After offering at the third temple we have

$$3(3^2x - y(1 + 3)) - y = 3^3x - y(1 + 3 + 3^2)$$

flowers left.

After offering at the fourth temple we have:

$$3(3^3x - y(1 + 3 + 3^2)) - y = 3^4x - y(1 + 3 + 3^2 + 3^3)$$

flowers left.

But as before since we should have no flowers left, we get the equation:

$$3^4x - y(1 + 3 + 3^2 + 3^3) = 0. \quad (14)$$

By expressing coefficients as powers of 3 and sums of powers of 3, we see a geometric series hidden in the problem! You will see when we do the most general case that we actually will be left with no choice but to write coefficients in index form and as a sum of a geometric series.

To understand and solve equation (14) we need to do some rearranging and use the formula for the sum of the geometric series

$$1 + 3 + 3^2 + 3^3 = \left(\frac{3^4 - 1}{3 - 1} \right).$$

Rearranging we get

$$3^4x = y \left(1 + 3 + 3^2 + 3^3 \right) = y \left(\frac{3^4 - 1}{3 - 1} \right),$$

hence $2(3^4x) = 3^4y - y$, and so,

$$y = 3^4(y - 2x).$$

The equation $y = 3^4(y - 2x)$, which can be expressed as $40y - 81x = 0$ is again a linear Diophantine equation and can be solved as before, but keeping it in the form $y = 3^4(y - 2x)$, reveals much more!

The infinite solutions of equation (14) can be manipulated by setting different values for $y - 2x$. For example, if we let $y - 2x = 1$, we get $y = 3^4 = 81$ and $x = 40$ the smallest positive integer solution for the tripling problem. Setting $y - 2x = k$, where k is a positive integer will produce all the infinite positive integer solutions we want!

For the original problem, using the sum of geometric series, we would have got $y = 2^4(y - x)$ and again setting $y - x = k$, where k is a positive integer, produces all the solutions we want!

I think you will agree with me that when we use geometric series we can see the connection between x , y , n and m .

Let the temples and lotuses go forth and multiply

Now let us attack the case when there are n temples and the flowers increase m fold.

As before after offering at the first temple we have

$$mx - y$$

flowers left.

And after offering at the second temple we have

$$m(mx - y) - y = m^2x - my - y = m^2x - (1 + m)y$$

flowers left.

After offering at the third temple we have:

$$m(m^2x - (m + 1)y) - y = m^3x - (m^2 + m + 1)y$$

flowers left.

Following the patterns after offering at the n^{th} temple we have

$$m^n x - (1 + m + m^2 + \cdots + m^{n-1})y$$

flowers left. Since we should have no flowers left, we get the equation:

$$m^n x - (1 + m + m^2 + \cdots + m^{n-1})y = 0.$$

Using the formula for the sum of the geometric series

$$1 + m + m^2 + \cdots + m^{n-1} = \left(\frac{m^n - 1}{m - 1} \right)$$

and rearranging as before we get

$$y = m^n [y - (m - 1)x]. \quad (15)$$

Once again setting $y - (m - 1)x = k$, where k is a positive integer, produces the infinite solutions we desire.

We can now please as many goddesses with as many lotuses as they wish. Beware, they all should have the same number though!

About the author

SHASHIDHAR JAGADEESHAN has been teaching mathematics for over 35 years. He is a firm believer that mathematics is a human endeavour, and his interest lies in conveying the beauty of mathematics to students and demonstrating that it is possible to create learning environments where children enjoy learning mathematics.

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Proof-Based Grid-Puzzles 3: Pentagon on a Square Grid

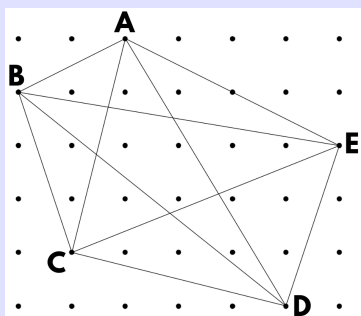
Article contributed by Rossi D'Souza

Introduction

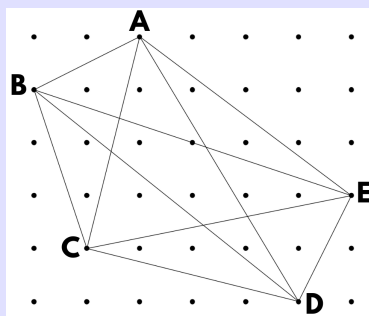
The “Pentagon on a Square Grid” is a hands-on activity that can be carried out with middle to high-school students with either a grid board and rubber-bands, or with a graph paper, a pencil and a ruler. The puzzle is as follows:

On a square grid, construct a five-pointed star (it will inevitably be an irregular pentagram) such that each vertex is a grid point. Now join these 5 points to form a pentagon. Now that we have 10 line segments, prove that the midpoint of at least one line segment is a grid point.

For example, in the grid pentagon cum pentagram star $ABCDE$ in Figure 48 (a), the midpoint of line segment EA is a grid point. When vertex E is shifted one unit down, the mid-point of BE falls on a grid point as seen in Figure 48 (b).



(a)



(b)

Figure 48: Grid Pentagon and star

While carrying out this activity with students, ask them to notice that no matter which grid points are used to construct the pentagon and star, at least one line segment — whether of the star or of the pentagon — will have its midpoint at a grid point.

In order to prove this observation, we invoke the Pigeonhole Principle which we shall first discuss briefly with the help of some simple problems.

The proof of this activity also serves as a simplified introduction into the Pigeonhole Principle which will have wide applications in higher mathematics as well as computer science.

The Pigeonhole Principle

You might have encountered the following riddle.

A box contains a large number of identically shaped red socks and blue socks, all mixed up. How many socks would you need to pick out at random to make sure that you have a pair of same coloured socks?

The answer as you rightly guessed is 3. If you pick two socks, they may be of different colours. But a third sock will guarantee that at least one colour is repeated, thereby giving a pair. Indeed, it is also possible that all three socks are of the same colour.

We can extend this problem to a box containing a mixture of three different coloured socks, in which case, you would need to pick out 4 socks to ensure that one colour is expressed as a pair.

Now let's look at a mathematical problem that follows a similar logic:

Given any four positive integers, prove that at least two of them leave the same remainder when divided by 3.

Here too the solution is simple, and follows similar reasoning to the sock-box problem: if you pick any four positive integers, given that their remainders when divided by 3 are either 0, 1 or 2, at least two of the integers will have the same remainder.

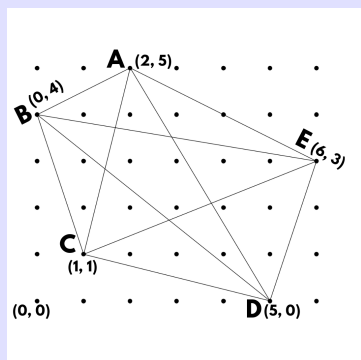
The above problems follow a pattern that can be expressed as follows:

Given a number of objects that need to be placed in a certain number of containers, if the number of objects is larger than the number of containers, then at least one container will contain more than one object.

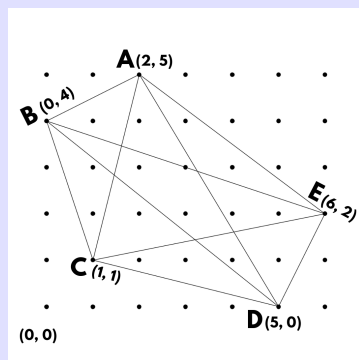
This statement gives the gist of the “Pigeonhole Principle” which is usually framed in terms of ‘pigeons’ occupying a set of ‘pigeonholes’ — if the pigeons outnumber the pigeonholes, then at least one pigeonhole will contain more than one pigeon.

Proof of the Grid-Pentagon-Pentagram Puzzle

To prove our observation that the midpoint of at least one line segment of our grid pentagon-and-star, we first assign integer coordinates to each grid point. For example, the coordinates of the pentagon in Figure 49 (a) are $(2,5)$, $(0,4)$, $(1,1)$, $(5,0)$, $(6,3)$.



(a)



(b)

Figure 49: Grid Pentagon and star with coordinates

Now consider 4 pigeonholes defined as follows: The first pigeonhole comprises all points whose x - & y -coordinates are both even. The second

pigeonhole has x -coordinate even and y -coordinate odd. The third pigeonhole has x -coordinate odd and y -coordinate even. And the fourth pigeonhole has both coordinates odd.

For example, vertex B of Figure 49 (a) will occupy the first pigeonhole as both its coordinates are even. Similarly vertex D will occupy the third pigeonhole as its x -coordinate is odd and y -coordinate is even.

If two points belong to the same pigeonhole, their corresponding x - & y -coordinates will have the same parity. For example, vertices $A(2,5)$ and $E(6,3)$ have x -coordinate even and y -coordinate odd, and thus occupy the same pigeonhole.

The coordinates of the midpoint of a line segment whose end points lie in the same pigeonhole will be the arithmetic mean of two odd or two even numbers, each of which is an integer. Thus the midpoint will be a grid point.

For example, suppose that vertex E is moved 1 unit down from $(6,3)$ to $(6,2)$. It will then share the pigeonhole with vertex $B(0,4)$ whose coordinates are both even. Consequently, the midpoint of BE will be a grid point.

In our problem we have 5 points connected to each other with line segments. Regarding the 5 points as pigeons, at least two of them will lie in the same pigeonhole (i.e., their x -coordinates and y -coordinates respectively will have the same parity). The midpoint of the line segment formed by joining those two points will necessarily be a grid point. This completes our proof. ■

Proof-based puzzle activities

This article is the third in a series on Proof-Based Grid Puzzles. The puzzles—The Hopping Horse Puzzle and the Grid-Diagonal Puzzle—presented in the previous two articles, [1] and [2], while also based on a square grid, used different mathematical principles, demanded different approaches in their analyses, and required different theorem-proving strategies. The Hopping Horse Puzzle involved principles from Graph Theory; the Grid-Diagonal Puzzle was based on similar triangles and the use of coordinate geometry; and the Grid-Pentagon-Pentagram Puzzle used the Pigeonhole Principle.

Incorporating puzzle-analysis in mathematics teaching highlights the need for an imaginative approach towards theorem-proving and problem-solving — given the very nature of mathematics as a creative social practice.

Acknowledgements

I thank Praneta Jha for the mathematical discussions that went into the development of this geometry activity as well as for editing this manuscript.

Further reading

- [1] D'Souza, R. (2024). Proof-Based Grid-Puzzles – 1: The Hopping Horse Puzzle. *The Compass* 1(2), 78–86.
- [2] D'Souza, R. (2025). Proof-Based Grid-Puzzles – 2: The Grid-Diagonal Puzzle. *The Compass* 2(1), 61–67.

About the author

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An Algorithm for Finding Approximate Square Roots

Here is a remarkably compact algorithm to find the approximate square root of a number. It yields the answer in a single ‘pass’ (i.e., no iterations are required), and its accuracy level can be as high as one wants.

Here are the steps of the algorithm. Let x be the given (positive) number whose square root is required. We shall assume that $x > 1$.

1. Locate a number close to x which is a perfect square. Denote it by a^2 . (Not that a does not have to be an integer. But $a \geq 1$.)
2. Then the following number y is an approximation for $\sqrt{x}$:

$$y = \frac{x + a^2}{2a}. \quad (16)$$

And that’s it — as simple as that!

Let’s look at some examples to illustrate how it works.

Example 1. *The square root of 2.* Let $x = 2$. Now let us choose a number close to x for which we have an exact square root. Let's choose $a^2 = 2.25$, i.e., $a = 1.5$. Here's the approximation we get:

$$\sqrt{2} \approx \frac{2 + 2.25}{2 \times 1.5} = \frac{4.25}{3} = \frac{17}{12} \approx 1.4167.$$

We have obtained an approximation which isn't too bad.

But we can do better than that! Let's choose $a^2 = 1.96$, i.e., $a = 1.4$. This yields:

$$\sqrt{2} \approx \frac{2 + 1.96}{2 \times 1.4} = \frac{3.96}{2.8} = \frac{99}{70} \approx 1.41429.$$

This approximation may be compared with

$$\sqrt{2} = 1.41421\ 35623\ 7309\dots$$

Not bad at all.

Example 2. *The square root of 3.* Let $x = 3$. Let's choose $a^2 = 2.89$, i.e., $a = 1.7$. Here's the approximation we get:

$$\sqrt{3} \approx \frac{3 + 2.89}{2 \times 1.7} = \frac{5.89}{3.4} = \frac{589}{340} \approx 1.7323.$$

This approximation may be compared with

$$\sqrt{3} = 1.73205\ 08075\ 6887\dots$$

We can do better than this by choosing a appropriately.

For example, let us choose $a^2 = \frac{49}{16} = 3.0625$, with $a = \frac{7}{4} = 1.75$. This yields

$$\sqrt{3} \approx \frac{3 + 3.0625}{2 \times 1.75} = \frac{6.0625}{3.5} = \frac{97}{56} \approx 1.7321.$$

Much better!

Example 3. *The square root of 5.* Let $x = 5$. Let's choose $a^2 = 4.84$, i.e., $a = 2.2$.

Here's the approximation we get:

$$\sqrt{5} \approx \frac{5 + 4.84}{2 \times 2.2} = \frac{9.84}{4.4} = \frac{123}{55} \approx 2.2364.$$

With $a^2 = 2.3^2 = 5.29$ we get:

$$\sqrt{5} \approx \frac{5 + 5.29}{2 \times 2.3} = \frac{10.29}{4.6} = \frac{1029}{460} \approx 2.237.$$

This approximation may be compared with

$$\sqrt{5} = 2.23606\ 79774\ 9978\dots$$

Explanation

The method cries out for an explanation! Here is one way of understanding the underlying mechanism.

Let us write

$$x = a^2 + h, \tag{17}$$

where $|h|$ is small compared with a . (We generally write this symbolically as $|h| \ll a$.) Then we have,

$$\begin{aligned} \sqrt{x} &= \sqrt{a^2 + h} \\ &= a \cdot \sqrt{1 + \frac{h}{a^2}} \\ &= a \cdot \left(1 + \frac{h}{2a^2} - \frac{h^2}{8a^4} + \dots \right), \quad \text{via the binomial theorem} \\ &= a + \frac{h}{2a} - \frac{h^2}{8a^3} + \dots \end{aligned}$$

As we have assumed that $|h| \ll a$, it seems reasonable and permissible to neglect h^2 in comparison with h , and therefore, $\frac{h^2}{8a^3}$ in comparison with $\frac{h}{2a}$.

The above relation then yields the following approximation:

$$\sqrt{x} \approx a + \frac{h}{2a}.$$

This may be written as

$$\sqrt{x} \approx \frac{2a^2 + h}{2a} = \frac{(a^2 + h) + a^2}{2a},$$

in other words,

$$\sqrt{x} \approx \frac{x + \text{close-by perfect square}}{2 \times \text{square root of close-by perfect square}}. \quad (18)$$

Similar approximate formulas may be found for extraction of cube roots and fourth roots.

Article contributed by the editors

2025, Revisited – Part 1

In the January 2025 issue, Prithwiji De highlighted an interesting property of number 2025,

$$2025 = (20 + 25)^2,$$

and asked about other four-digit numbers that have such a property. It turns out that there are just two other such numbers:

$$3025 = (30 + 25)^2 \quad \text{and} \quad 9801 = (98 + 01)^2.$$

It would be of interest to extend this exploration further. What follows is an account of two such extensions.

Extension to five-digit and six-digit numbers

For the case of four digits, we need to solve the following problem: Find two-digit numbers a, b such that

$$100a + b = (a + b)^2.$$

A careful search reveals that there are three solutions to this equation, as noted above.

Moving to five-digit and six-digit numbers, the problem we have is this: Find all pairs of two-digit and three-digit numbers a, b such that

$$1000a + b = (a + b)^2. \quad (19)$$

We could do so by “brute force” using a fast computer to search through all the possibilities. That will surely yield the result, but it will not increase our understanding in any way. Let us avoid this approach. What follows, below, is an attempt to solve this problem analytically.

Let’s first treat (19) as an equation in a and solve for a . Simplifying the equation, we obtain,

$$a^2 + 2a(b - 500) + b(b - 1) = 0. \quad (20)$$

This is a quadratic equation in a . Its discriminant is

$$2^2(b - 500)^2 - 4b(b - 1) = 4(250000 - 999b). \quad (21)$$

For equation (19) to have integer solutions, the discriminant, given by (21) must be a perfect square. (We do not know if it will be sufficient.)

Now, if $4(250000 - 999b)$ is a perfect square, so is $250000 - 999b$. So, let us first ask: **For which integer values of b is $250000 - 999b$ a perfect square?**

This turns out to be a very interesting question. Obviously, we can answer this too using brute force search. But we shall avoid this approach.

Let

$$250000 - 999b = x^2, \quad (22)$$

where x is a non-negative integer. Then we have, $250000 - x^2 = 999b$, so,

$$(500 - x)(500 + x) = 999b. \quad (23)$$

We must find integers x and b that solve equation (23). We argue as follows.

- Since $500 - x \geq 0$, we have $x \leq 500$.
- The prime factorization of 999 is $999 = 3^3 \times 37$. So, let us rewrite equation (23) as,

$$(500 - x)(500 + x) = 3^3 \times 27 \times b. \quad (24)$$

Each prime divisor on the right side (namely, 3, 37, and the primes that divide b) must be a divisor of either $500 - x$ or $500 + x$ or both.

- The sum of the two factors on the left side is 1000, which is not divisible by either 3 or 37. This tells us that 3 (and therefore 27 too) is a divisor of just *one* of $500 - x$ and $500 + x$; and likewise for 37. (For, if 3 were a divisor of both $500 - x$ and $500 + x$, then it would divide their sum as well, i.e., it would divide 1000; but it does not. The same logic works for 37.)

We therefore consider the following possibilities.

- 27 and 37 are both divisors of $500 + x$. This quickly leads to a solution, as we show below.
- 27 and 37 are both divisors of $500 - x$. But since $27 \times 37 = 999$, it is not possible for 27 and 37 to both be divisors of $500 - x$. So this possibility can be ruled out.
- 27 is a divisor of $500 - x$ and 37 is a divisor of $500 + x$.
- 37 is a divisor of $500 - x$ and 27 is a divisor of $500 + x$.

The first possibility leads to a quick solution. If $500 + x$ is divisible by both 27 and 37, then it is divisible by $27 \times 37 = 999$. But $500 + x \leq 1000$. The only multiple of 999 between 500 and 1000 is 999 itself. Hence $500 + x = 999$,

giving $x = 499$ and $b = 1$. Referring back to (19) we get,

$$1000a + 1 = (a + 1)^2, \quad \therefore 998a = a^2, \quad \therefore a = 998.$$

(The solution $a = 0$ is not meaningful here, so we have not listed it.) This corresponds to the solution

$$(998 + 001)^2 = 998001.$$

Now let us consider the other two possibilities. In both cases we have a multiple of 27 and a multiple of 37 that add up to 1000. Let us therefore look for non-negative integers m, n such that

$$27m + 37n = 1000. \tag{25}$$

This is an example of a *linear Diophantine equation* — a linear equation with more than one unknown, with the requirement that all the unknowns must take non-negative integer values. As it has just two unknowns, there is a standard way of finding its solutions.

We first note that since 27 and 37 are coprime, integers u, v can be found such that

$$27u + 37v = 1. \tag{26}$$

There are different ways of finding u and v . One way is to use Euclid's division algorithm to find the GCD:

$$37 = 1 \times 27 + 10,$$

$$27 = 2 \times 10 + 7,$$

$$10 = 1 \times 7 + 3,$$

$$7 = 2 \times 3 + 1.$$

The '1' in the last line shows that the GCD is 1. We now work backwards:

$$\begin{aligned}
 1 &= 7 - 2 \times 3 &= 7 - 2(10 - 1 \times 7) \\
 &= 3 \times 7 - 2 \times 10 &= 3(27 - 2 \times 10) - 2 \times 10 \\
 &= 3 \times 27 - 8 \times 10 &= 3 \times 27 - 8(37 - 1 \times 27) \\
 &= 11 \times 27 - 8 \times 37.
 \end{aligned}$$

So, $27u + 37v = 1$ when $u = 11$ and $v = -8$. (Please check the computation.)

We now have a solution in integers of the equation $27u + 37v = 1000$,

$$27 \times 11000 + 37 \times (-8000) = 1000.$$

However, we require *solutions in non-negative integers*. To find these, we add to the above solution (i.e., $u = 11000, v = -8000$) suitable multiples of solutions to the equation $27u + 37v = 0$. Since 27 and 37 are coprime, the general solution in integers to the equation $27u + 37v = 0$ is

$$u = 37k, \quad v = -27k,$$

where k is an arbitrary integer. Hence the general solution in integers to the equation $27m + 37n = 1000$ is

$$m = 37k + 11000, \quad n = -27k - 8000, \tag{27}$$

where k is an arbitrary integer. Since m, n are non-negative we must have

$$37k + 11000 \geq 0, \quad \therefore k \geq -\frac{11000}{37}, \quad \therefore k \geq -297.3,$$

which means that $k \geq -297$ (since k is an integer); and

$$-27k - 8000 \geq 0, \quad \therefore k \leq -\frac{8000}{27}, \quad \therefore k \leq -296.3,$$

which means that $k \leq -297$ (since k is an integer). The two conditions together show that $k = -297$ (this is forced). Therefore there is just one solution in non-negative integers to the equation $27m + 37n = 1000$,

$$m = 37 \times (-297) + 11000 = 11,$$

$$n = -27 \times (-297) - 8000 = 19.$$

So we have found the respective multiples of 27 and 37 that add up to 1000; they are the numbers $27 \times 11 = 297$ and $37 \times 19 = 703$ respectively.

Referring back to the equation (24), we see that these values correspond to

$$x = 703 - 500 = 203, \quad b = 11 \times 19 = 209.$$

Having found b , we can find a by solving the quadratic equation

$$1000a + 209 = (a + 209)^2.$$

On opening the brackets and bringing all the terms to one side, we obtain

$$a^2 - 582a + 43472 = 0.$$

This turns out to have a nice factorization (which we find after a bit of trial and error),

$$(a - 88)(a - 494) = 0.$$

So, at the end of this long search, we have located all the five-digit and six-digit numbers that have the required property:

$$88209 = (88 + 209)^2,$$

$$494209 = (494 + 209)^2,$$

$$998001 = (998 + 001)^2.$$

And that's it!

Remark

The reader should go back to the four-digit case and find all the solutions (we know that there are just three of them) using this approach.

There is one other extension possible, which is to search for solutions in the set of rational numbers; that is, we relax the condition that the variables must assume integer values. This problem turns out to be of great interest but we shall explore it only in the next issue.

Article contributed by Shailesh Shirali

Heron's Formula for Area of a Triangle – 4

Article contributed by Uttara Surya

We continue the series of articles on Heron's formula for area of a triangle, and now take up the proof of Brahmagupta's theorem which gives a remarkable formula for the area of a cyclic quadrilateral.

Statement of the theorem

Theorem (Brahmagupta). *Let $ABCD$ be a cyclic quadrilateral. Let its side-lengths be $AB = a$, $BC = b$, $CD = c$, and $DA = d$, and let $s = \frac{1}{2}(a + b + c + d)$ be the semi-perimeter. Then the area K of the quadrilateral is given by*

$$K = \sqrt{(s - a)(s - b)(s - c)(s - d)}. \quad (28)$$

The formula is amazingly compact and symmetric. One can well imagine the pleasure that its discovery would have given Brahmagupta.

Trigonometric proof

There is a straightforward proof using high-school trigonometry — an approach familiar to most students. We now describe this proof in brief.

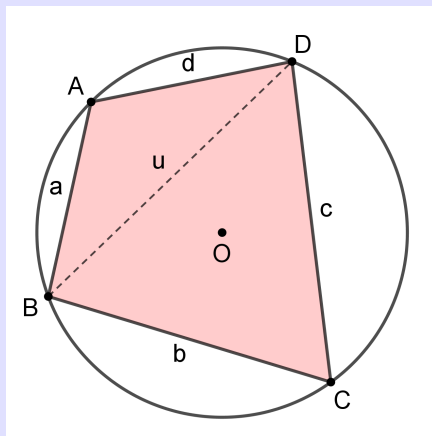


Figure 50: Formula for area of a cyclic quadrilateral

The configuration is depicted in Figure 50. Let u be the length of diagonal BD . Then we have, by the cosine rule,

$$\begin{cases} u^2 &= a^2 + d^2 - 2ad \cos A, \\ u^2 &= b^2 + c^2 - 2bc \cos C. \end{cases}$$

We also have $\cos C = -\cos A$, since $\angle A + \angle C = 180^\circ$. So,

$$\begin{cases} u^2 &= a^2 + d^2 - 2ad \cos A, \\ u^2 &= b^2 + c^2 + 2bc \cos A. \end{cases}$$

Subtracting, we get

$$2(ad + bc) \cos A = a^2 - b^2 - c^2 + d^2,$$

hence

$$\cos A = \frac{a^2 - b^2 - c^2 + d^2}{2(ad + bc)}. \quad (29)$$

This relation allows us to express $\sin A$ in terms of a, b, c, d . We have,

$$\begin{aligned} \sin^2 A &= 1 - \cos^2 A \\ &= 1 - \left(\frac{a^2 - b^2 - c^2 + d^2}{2(ad + bc)} \right)^2, \end{aligned}$$

$$\therefore 4(ad + bc)^2 \sin^2 A = 4(ad + bc)^2 - (a^2 - b^2 - c^2 + d^2)^2. \quad (30)$$

The expression on the right side of (30) can be factorized using the familiar difference-of-two-squares formula repeatedly:

$$\begin{aligned} &4(ad + bc)^2 - (a^2 - b^2 - c^2 + d^2)^2 \\ &= (2ad + 2bc + a^2 - b^2 - c^2 + d^2) \cdot (2ad + 2bc - a^2 + b^2 + c^2 - d^2) \\ &= ((a + d)^2 - (b - c)^2) \cdot ((b + c)^2 - (a - d)^2) \\ &= (a + d + b - c)(a + d - b + c)(b + c + a - d)(b + c - a + d) \\ &= (-a + b + c + d)(a - b + c + d)(a + b - c + d)(a + b + c - d) \\ &= 2(s - a) \cdot 2(s - b) \cdot 2(s - c) \cdot 2(s - d), \end{aligned} \quad (31)$$

where $s = (a + b + c + d)/2$.

It is interesting that the intimidating expression on the right hand side of (30) factorizes so neatly!

The familiar formula for area of a triangle, “half the product of two sides and the sine of the included angle,” now tells us that

$$\begin{aligned} K &= \frac{1}{2}ad \sin A + \frac{1}{2}bc \sin C \\ &= \frac{1}{2}(ad + bc) \sin A, \end{aligned}$$

since $\sin A = \sin C$ (as $\angle A + \angle C = 180^\circ$). Hence,

$$\begin{aligned} 16K^2 &= 4(ad + bc)^2 \sin^2 A \\ &= 4(ad + bc)^2 - (a^2 - b^2 - c^2 + d^2)^2 \quad \text{from (30)} \\ &= 2(s - a) \cdot 2(s - b) \cdot 2(s - c) \cdot 2(s - d) \quad \text{from (31)} \\ &= 16(s - a)(s - b)(s - c)(s - d), \end{aligned}$$

and so we arrive at the remarkable formula,

$$K = \sqrt{(s - a)(s - b)(s - c)(s - d)}.$$

We do not know the path taken by Brahmagupta in proving this formula. But it seems likely to us that his approach was more geometrical in nature. It was probably along the lines of the proof described below.

A proof using ‘pure geometry’

The proof described below has been adapted from *Yuktibhasa*, one of the great works of the Kerala School of mathematics.

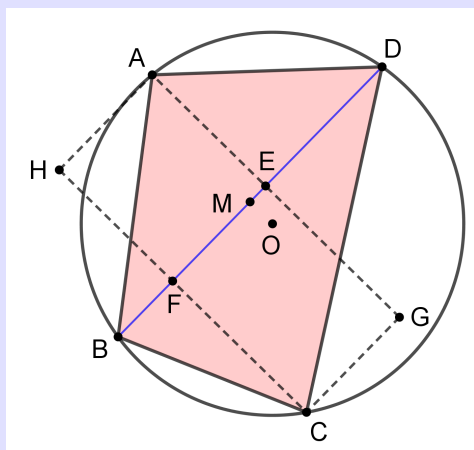


Figure 51: Formula for area of a cyclic quadrilateral — a geometric proof

Draw diagonal BD (see Figure 51), and drop perpendiculars from A and C to line BD ; the feet of these perpendiculars are, respectively, E and F (so $AE \perp BD$, $CF \perp BD$). Next, drop perpendiculars from A to line CF and from C to line AE ; the feet of these perpendiculars are, respectively, H and G (so $AH \perp CF$, $CG \perp AE$). Let M be the midpoint of BD .

Before proceeding let us explain the logic behind the construction. Clearly,

$$\begin{aligned}
 K &= \text{Area}(\triangle ABD) + \text{Area}(\triangle CBD) \\
 &= \frac{1}{2}BD \cdot AE + \frac{1}{2}BD \cdot CF \\
 &= \frac{1}{2}BD \cdot (AE + CF) = \frac{1}{2}BD \cdot HC,
 \end{aligned}$$

since $AHCG$ is a rectangle.

It follows that if we find an expression for the length of HC , then we will obtain a formula for the area of the quadrilateral.

And since $HC^2 = AC^2 - AH^2 = AC^2 - EF^2$, if we find an expression for the length of EF , we will obtain a formula for the area of the quadrilateral.

We now have,

$$\begin{aligned} AB^2 &= BE^2 + AE^2 \\ &= (BM \pm ME)^2 + AE^2, \\ AD^2 &= DE^2 + AE^2 \\ &= (DM \mp ME)^2 + AE^2. \end{aligned}$$

Here we use the signs $\pm$ and $\mp$ because we do not know in advance where M is located relative to E and F .

By subtraction, we get,

$$\begin{aligned} |AB^2 - AD^2| &= |(BM \pm ME)^2 - (DM \mp ME)^2| \\ &= |4BM \cdot ME| \quad (\text{since } BM = DM) \\ &= 2BD \cdot |ME|. \end{aligned}$$

From this it follows that

$$|ME| = \frac{|AB^2 - AD^2|}{2BD}. \quad (32)$$

In the same way we get

$$|MF| = \frac{|CD^2 - CB^2|}{2BD}. \quad (33)$$

From (32) and (33) we get, by addition,

$$EF = \frac{|(AB^2 + CD^2) - (BC^2 + AD^2)|}{2BD}. \quad (34)$$

Now we are in a position to find the length of HC :

$$\begin{aligned} HC^2 &= AC^2 - AH^2 \\ &= AC^2 - EF^2 \\ &= AC^2 - \left(\frac{|(AB^2 + CD^2) - (BC^2 + AD^2)|}{2BD} \right)^2. \end{aligned} \quad (35)$$

Hence,

$$\begin{aligned} K^2 &= \frac{1}{4}BD^2 \cdot HC^2 \\ &= \frac{AC^2 \cdot BD^2}{4} - \frac{(AB^2 + CD^2 - BC^2 - AD^2)^2}{16} \end{aligned} \quad (36)$$

We know from Ptolemy's theorem (more on this, below) that

$$AC \cdot BD = AB \cdot CD + BC \cdot AD. \quad (37)$$

We therefore have,

$$\begin{aligned} K^2 &= \frac{1}{4}(ac + bd)^2 - \frac{(a^2 - b^2 + c^2 - d^2)^2}{16} \\ &= \frac{1}{16} \left(4(ac + bd)^2 - (a^2 - b^2 + c^2 - d^2)^2 \right) \\ &= \frac{1}{16} \left((2ac + 2bd)^2 - (a^2 - b^2 + c^2 - d^2)^2 \right). \end{aligned}$$

The expression looks intimidating, just like (30) did, but as earlier it is only a difference of two squares. So, it is not difficult to factorize it:

$$\begin{aligned}
 16K^2 &= (2ac + 2bd)^2 - (a^2 - b^2 + c^2 - d^2)^2 \\
 &= (2ac + 2bd + a^2 - b^2 + c^2 - d^2) \cdot (2ac + 2bd - a^2 + b^2 - c^2 + d^2) \\
 &= ((a + c)^2 - (b - d)^2) \cdot ((b + d)^2 - (a - c)^2) \\
 &= (a + c + b - d) \cdot (a + c - b + d) \cdot (b + d + a - c) \cdot (b + d - a + c) \\
 &= (b + c + d - a)(c + d + a - b)(d + a + b - c)(a + b + c - d) \\
 &= 16(s - a)(s - b)(s - c)(s - d), \quad \text{where } s = \frac{1}{2}(a + b + c + d).
 \end{aligned}$$

Thus we arrive at Brahmagupta's beautiful formula for the area of a cyclic quadrilateral with sides a, b, c, d and semi-perimeter s :

$$K = \sqrt{(s - a)(s - b)(s - c)(s - d)}. \quad (38)$$

There is another proof that too is geometric in nature and avoids the use of trigonometric expressions. We describe this in the next article of this series. We also describe (but without proof) a remarkable formula that extends Brahmagupta's theorem to general convex quadrilaterals.

Appendix: Ptolemy's theorem

To close this article we quote the very important result known as *Ptolemy's theorem*.

Theorem (Ptolemy). *Let $ABCD$ be a cyclic quadrilateral. Then the lengths of the sides and the diagonals obey the following equality:*

$$AC \cdot BD = AB \cdot CD + AD \cdot BC. \quad (39)$$

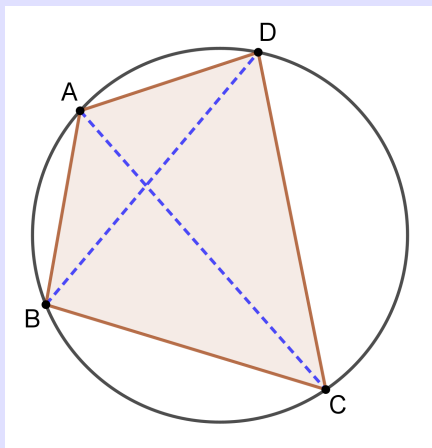


Figure 52: Ptolemy's theorem: $AC \cdot BD = AB \cdot CD + AD \cdot BC$

We shall study the proof of the theorem and some of its applications in the next issue.

Further reading

[1] Wikipedia, "Heron's formula."

https://en.wikipedia.org/wiki/Heron's_formula

- [2] Shannon Umberger, “A Geometric Proof of Heron’s Formula.”
<https://jwilson.coe.uga.edu/EMT668/EMAT6680.2000/Umbberger/MATH7200/HeronFormulaProject/GeometricProof/geoproof.html>
- [3] Wikipedia, “Ptolemy’s theorem.”
https://en.wikipedia.org/wiki/Ptolemy's_theorem
- [4] Jim Wilson (University of Georgia), “Ptolemy’s theorem.” <https://jwilson.coe.uga.edu/emt725/Ptolemy/Ptolemy.html>
- [5] Howard Eves, “An Introduction to the History of Mathematics” (1969). Holt, Rinehart and Winston. (Chapter 6, page 172)

About the author

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Surprising limits – 2

In the previous issue we discussed examples of the indeterminate form $\infty - \infty$, all involving the functions $\frac{1}{x}$ and $\frac{1}{\sin x}$ in the neighbourhood of $x = 0$. On each occasion the limit surprised us. We look at two more examples of this kind in this issue.

A limit involving the sine function

We consider the following limit:

$$L = \lim_{x \rightarrow 0} (1 + \sin x)^{1/x}. \quad (40)$$

As always in situations where our intuition can lead us astray, it helps to do some number crunching. Here is a list of values taken by the function $g(x) = (1 + \sin x)^{1/x}$ for $x = 0.1, 0.01, \dots, 0.00001$:

x	10^{-1}	10^{-2}	10^{-3}	10^{-4}	10^{-5}
$g(x)$	2.59	2.70	2.717	2.718	2.7183

The numerical evidence seems very clear. It looks as though $L = e$.

Here is one way of solving the problem. We assume (to start with) that x approaches zero from the positive side, i.e., $x \rightarrow 0^+$. Let $n = \frac{1}{x}$. Then $x \rightarrow 0^+$ is equivalent to $n \rightarrow +\infty$ (we shall write this as just ∞). So we must find the following limit:

$$L = \lim_{n \rightarrow \infty} \left(1 + \sin \left(\frac{1}{n} \right) \right)^n. \quad (41)$$

Making use of the fact that for $t \approx 0$,

$$\sin t = t - \frac{t^3}{3!} + \cdots,$$

we see that L is equal to the following:

$$\begin{aligned} &= \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} - \frac{1}{6n^3} + \cdots \right)^n \\ &= \lim_{n \rightarrow \infty} \left(1 + n \left(\frac{1}{n} - \frac{1}{6n^3} + \cdots \right) + \frac{n(n-1)}{2} \left(\frac{1}{n} - \frac{1}{6n^3} + \cdots \right)^2 + \cdots \right) \\ &= \lim_{n \rightarrow \infty} \left(1 + \left(1 - \frac{1}{6n^2} + \cdots \right) + \frac{n(n-1)}{2} \left(\frac{1}{n^2} - \frac{1}{3n^4} + \cdots \right) + \cdots \right) \\ &= \lim_{n \rightarrow \infty} \left(1 + 1 + \frac{1 - 1/n}{2} + \cdots + \left(\text{terms involving } \frac{1}{n^2}, \frac{1}{n^4}, \text{ etc} \right) \right) \\ &= 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \cdots = e. \end{aligned}$$

Hence the required answer is e . (Not a great surprise!) ■

Another approach: Here is another way of arriving at the same answer. We use the following result:

$$\lim_{x \rightarrow p} (f(x))^{g(x)} = \left(\lim_{x \rightarrow p} f(x) \right)^{\left(\lim_{x \rightarrow p} g(x) \right)},$$

provided that

$$\lim_{x \rightarrow p} f(x) > 0,$$

or

$$\lim_{x \rightarrow p} f(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow p} g(x) \text{ is positive and finite.}$$

Assuming this result, we reason as follows.

$$\begin{aligned} \lim_{x \rightarrow 0} (1 + \sin x)^{1/x} &= \lim_{x \rightarrow 0} \left((1 + \sin x)^{1/\sin x} \right)^{\sin x/x} \\ &= A^B, \end{aligned}$$

where

$$A = \lim_{x \rightarrow 0} (1 + \sin x)^{1/\sin x} = \lim_{t \rightarrow 0} (1 + t)^{1/t}$$

$$B = \lim_{x \rightarrow 0} \frac{\sin x}{x}.$$

It is well-known that $A = e$ and $B = 1$. Hence $L = e^1$, i.e.,

$$L = e. \quad \blacksquare$$

A limit involving the tan function

We are going to find the following limits:

$$\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{\tan x} \right) \quad (42)$$

and

$$\lim_{x \rightarrow 0} \left(\frac{1}{x^2} - \frac{1}{\tan^2 x} \right). \quad (43)$$

Let

$$f(x) = \frac{1}{x} - \frac{1}{\tan x}, \quad h(x) = \frac{1}{x^2} - \frac{1}{\tan^2 x}.$$

Let us compute a table of values of $f(x)$ and $h(x)$ for small values of x .

x	10^{-1}	10^{-2}	10^{-3}	10^{-4}	10^{-5}
$f(x)$	0.0334	0.00333	0.000333	0.0000333	0.00000333
$h(x)$	0.666	0.667	0.6667	0.66667	0.666667

The numerical evidence seems clear. It looks as though

$$\lim_{x \rightarrow 0} f(x) = 0, \quad \lim_{x \rightarrow 0} h(x) = \frac{2}{3}.$$

Evaluation of the limits Let us find the limits using the approach followed in the previous issue. We start by finding an expression that approximates the value of $\tan x$ for $x \approx 0$. We have,

$$\begin{aligned} \tan x &= \frac{\sin x}{\cos x} \approx \frac{x - x^3/6}{1 - x^2/2} \\ &\approx \left(x - \frac{x^3}{6}\right) \left(1 + \frac{x^2}{2}\right) = x - \frac{x^3}{6} + \frac{x^3}{2} - \frac{x^5}{12} \\ &\approx x + \frac{x^3}{3} \quad (\text{discarding terms with degree larger than 3}). \end{aligned}$$

Hence,

$$\begin{aligned}\frac{1}{\tan x} &\approx \frac{1}{x + x^3/3} \\ &= \frac{1}{x} \cdot \frac{1}{1 + x^2/3} \\ &\approx \frac{1}{x} \cdot \left(1 - \frac{x^2}{3}\right),\end{aligned}$$

and so, for $x \approx 0$,

$$\frac{1}{\tan x} \approx \frac{1}{x} - \frac{x}{3} \quad (44)$$

It follows that for $x \approx 0$,

$$\frac{1}{x} - \frac{1}{\tan x} \approx \frac{x}{3}. \quad (45)$$

The relation obtained above exactly agrees with the table of values that we computed earlier.

Next, we have,

$$\begin{aligned}\frac{1}{\tan^2 x} &\approx \frac{1}{x^2} \cdot \left(1 - \frac{x^2}{3}\right)^2 \\ &\approx \frac{1}{x^2} \cdot \left(1 - \frac{2x^2}{3}\right) \\ &= \frac{1}{x^2} - \frac{2}{3}.\end{aligned}$$

Hence, for $x \approx 0$,

$$\frac{1}{x^2} - \frac{1}{\tan^2 x} \approx \frac{2}{3}. \quad (46)$$

This is quite a surprising result but it too agrees exactly with the table of values that we computed earlier. ■

Article contributed by the editors

Two New Proofs of the Pythagorean Theorem

In the January 2025 issue of *The Compass*, we described a remarkable proof of the Pythagorean Theorem given by two school students, Calcea Johnson and Ne’Kiya Jackson. (See [2] and [1].) The novel aspect of the proof is its use of trigonometry. Now the same two students have come out with more proofs of the theorem (see [3]); indeed, their article is provocatively titled “Five or Ten New Proofs of the Pythagorean Theorem!” We present two of these proofs here.

Prerequisites

Before presenting the proofs we list the basic results from trigonometry that we need. As pointed out in the earlier article, we must ensure that we have proofs for these results that do not depend on the Pythagoras theorem.

We provide the necessary justifications in the appendix.

- (a) Definitions of sine and cosine, based on the right-angled triangle. (Note that this means that the functions are defined only for acute angles.)
- (b) The *addition formulas* for sine and cosine: if x, y are angles such that $0 < y < x < 90^\circ$ and $x + y < 90^\circ$, then

$$\sin(x + y) = \sin x \cos y + \cos x \sin y, \quad (47)$$

$$\sin(x - y) = \sin x \cos y - \cos x \sin y. \quad (48)$$

- (c) The *double angle formulas*: If x is an acute angle such that $0 < 2x < 90^\circ$, then

$$\sin 2x = 2 \sin x \cos x, \quad (49)$$

$$\cos 2x = \cos^2 x - \sin^2 x. \quad (50)$$

- (d) The *sine rule* (also called the ‘law of sines’): In $\triangle ABC$ (using standard notation), we have:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}. \quad (51)$$

The first of the “five new proofs” is the one presented in the January 2025 issue, so we start with the second proof.

In both these proofs, ABC is a given right-angled triangle, with $\angle C = 90^\circ$ and $\angle A < \angle B$; or, in terms of sides, $BC < AC$. Obviously, this means that the triangle is not isosceles. For reasons that will become clear during the course of the proofs, we need to consider the isosceles case separately.

The second proof

Given the right-angled $\triangle ABC$, we locate point D on BC extended so that AB is the internal bisector of angle CAD (Figure 53).

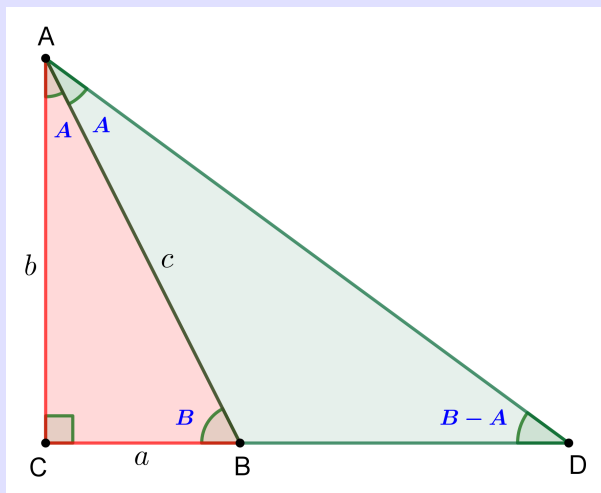


Figure 53: Second proof of the PT. AB internally bisects $\angle CAD$.

We will work out an expression for the length of BD in two different ways. Pythagoras theorem will follow immediately.

Using the sine rule we get,

$$\frac{CD}{\sin 2A} = \frac{AC}{\sin(B - A)}. \quad (52)$$

Now we have, using the addition and double-angle formulas,

$$\sin 2A = 2 \sin A \cos A = \frac{2ab}{c^2},$$

$$\sin(B - A) = \sin B \cos A - \cos B \sin A = \frac{b^2 - a^2}{c^2}.$$

Hence,

$$\frac{CD}{2ab/c^2} = \frac{AC}{(b^2 - a^2)/c^2},$$

giving,

$$CD = \frac{2ab}{c^2} \cdot \frac{bc^2}{b^2 - a^2} = \frac{2ab^2}{b^2 - a^2},$$

$$\begin{aligned} \therefore BD &= CD - BC = \frac{2ab^2}{b^2 - a^2} - a \\ &= \frac{a^3 + ab^2}{b^2 - a^2} = \frac{a(a^2 + b^2)}{b^2 - a^2}. \end{aligned} \quad (53)$$

We have obtained an expression for BD via $BD = CD - CB$. Let us obtain a second expression for the same length using the sine rule in $\triangle ABD$:

$$\begin{aligned} \frac{BD}{\sin A} &= \frac{AB}{\sin(B - A)}, \\ \therefore \frac{BD}{a/c} &= \frac{c}{(b^2 - a^2)/c^2}, \\ \therefore BD &= \frac{a}{c} \cdot \frac{c^3}{b^2 - a^2} = \frac{ac^2}{b^2 - a^2}. \end{aligned} \quad (54)$$

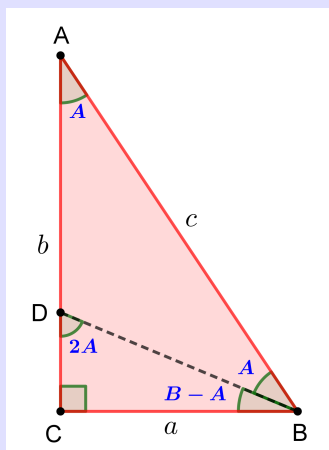
Comparing (53) and (54), we immediately obtain

$$c^2 = a^2 + b^2, \quad (55)$$

as required. ■

Comment We see how crucially dependent the above proof is on the addition formula and the double-angle formula. We will surely have to make good on our promise to provide proofs of these formulas that do not depend in any way on Pythagoras Theorem!

The third proof



Locate a point D on side AC such that $\angle ABD = \angle A$. Note that this will make $\angle ABD = \angle BAD$, $DA = DB$, and $\angle DBC = B - A$.

A simple way of locating D is to draw the perpendicular bisector of side AB ; then D is the point where this intersects AC .

We now compute the length of BD in two different ways.

Figure 54

We first note that,

$$\begin{aligned}\sin \angle BDC &= \frac{BC}{BD}, & \therefore \sin 2A &= \frac{a}{BD}, \\ \therefore \frac{2ab}{c^2} &= \frac{a}{BD},\end{aligned}$$

and therefore,

$$BD = \frac{c^2}{2b}. \quad (56)$$

From (56) we obtain,

$$CD = BD \sin(B - A) = \frac{c^2}{2b} \cdot \frac{b^2 - a^2}{c^2} = \frac{b^2 - a^2}{2b}. \quad (57)$$

Hence,

$$AD = AC - CD = b - \frac{b^2 - a^2}{2b} = \frac{a^2 + b^2}{2b}, \quad (58)$$

and since $BD = AD$, it follows that

$$BD = \frac{a^2 + b^2}{2b}. \quad (59)$$

From (56) and (59) we obtain,

$$c^2 = a^2 + b^2,$$

as required. ■

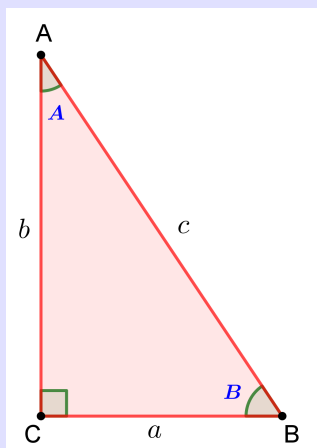
Query: Do you see now why we had to specify at the start that $\triangle ABC$ is not isosceles? This being so, the isosceles case has to be dealt with separately. We leave this task to the reader.

Appendix: A few justifications

We must now present justifications for what we had listed as ‘prerequisites’.

- (a) Definitions of sine and cosine.
- (b) The addition formulas for sine and cosine.
- (c) The double-angle formulas.
- (d) The sine rule.

Definitions: Let $\triangle ABC$ be right-angled at C , so $\angle A + \angle B = 90^\circ$.



Then we have (see figure at the left),

$$\sin A = \frac{BC}{AB} = \frac{a}{c}$$

and

$$\cos A = \frac{AC}{AB} = \frac{b}{c}.$$

The values of $\sin A$ and $\cos A$ depend only on the shape of the triangle (i.e., on $\angle A$ and $\angle B$), and not on its scale (i.e., its absolute size). This follows from similarity geometry.

Figure 55: Definition of sine and cosine

For the addition formulas, we present an elegant and compact proof without words (Figure 56).

In keeping with the philosophy of a proof without words, we shall not provide any explanations! We shall only advise you to examine the figure carefully, identify all angles in the figure that are equal to x , y and $x + y$ respectively, and then convince yourself that the expressions for the lengths of the various segments are as stated in the figure. After you do this, the identities for $\sin(x + y)$ and $\cos(x + y)$ will follow immediately. After you have done this, we invite you to draw an appropriate figure that will justify the formulas for $\sin(x - y)$ and $\cos(x - y)$.

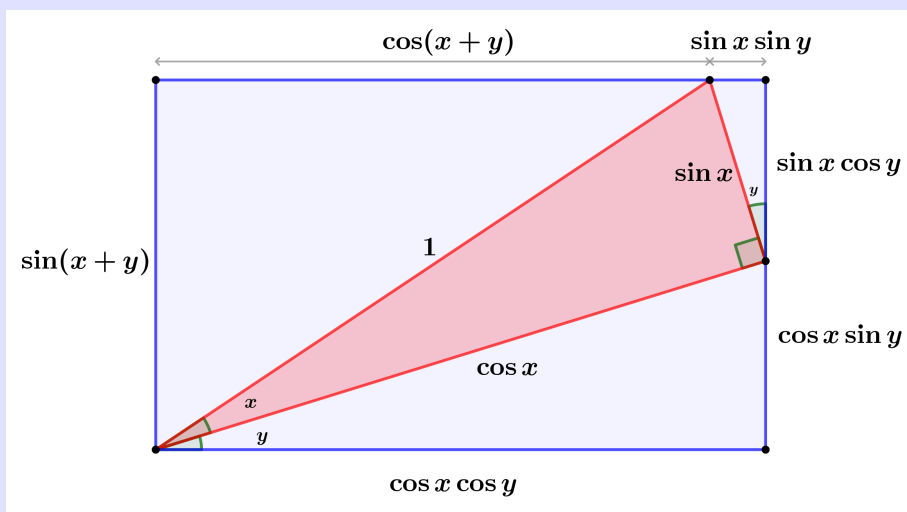


Figure 56: The addition formulas for sine and cosine follow just by looking closely at this figure!

For the double angle formulas we can use the addition formulas proved above. But we opt for a different proof based on area computation.

We consider an isosceles $\triangle ABC$ in which $AB = 1 = AC$, and $\angle BAC = 2x$. Let D be the midpoint of base BC ; then $AD \perp BC$. Using the definitions of sine and cosine, we get $AD = \cos x$ and $BD = \sin x = DC$, so $BC = 2 \sin x$. Therefore,

$$\text{Area of } \triangle ABC = \frac{1}{2} \cdot 2 \sin x \cdot \cos x = \sin x \cos x. \quad (60)$$

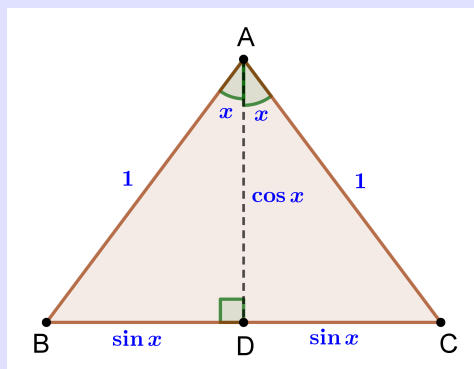


Figure 57

On the other hand we can use the formula “area of triangle equals half the product of two sides and the sine of the included angle.” This yields

$$\text{Area of } \triangle ABC = \frac{1}{2} \cdot 1 \cdot 1 \cdot \sin 2x = \frac{1}{2} \sin 2x. \quad (61)$$

From (60) and (61) we obtain the desired identity,

$$\sin 2x = 2 \sin x \cos x. \quad (62)$$

Closing remarks

As can be seen from the above proofs, the term phrase “trigonometric proof” holds less content than we may have supposed at the start. In reality, what underpins all these proofs is the use of results about similar triangles. It is more meaningful to say that a proof is based on similarity geometry than to describe it as a trigonometric proof.

Further reading

- [1] Shirali, Shailesh (2025), “A New Proof of the Theorem of Pythagoras.” *The Compass*, January 2025, pp. 139-149.
- [2] Shirali, Shailesh (2023), “Two New Proofs of the Pythagorean Theorem - Part I.” *At Right Angles*, July 2023. pp. 7-12. <https://publications.azimpremjiuniversity.edu.in/id/eprint/4877>
- [3] Ne’Kiya Jackson & Calcea Johnson (2024), “Five or Ten New Proofs of the Pythagorean Theorem,” *The American Mathematical Monthly*, November 2024, pp. 739-752. From <https://tinyurl.com/bddc7cks>.
- [4] Elisha Scott Loomis, *The Pythagorean Proposition*. From <https://files.eric.ed.gov/fulltext/ED037335.pdf>; see also <https://mathlair.allfunandgames.ca/pythprop.php>.
- [5] Wikipedia, “Pythagorean theorem.” From https://en.wikipedia.org/wiki/Pythagorean_theorem.
- [6] Cut-the-Knot, “Pythagorean Theorem.” From <https://www.cut-the-knot.org/pythagoras/>.
- [7] Leila Sloman, “2 High School Students Prove Pythagorean Theorem. Here’s What That Means.” From <https://www.scientificamerican.com/article/2-high-school-students-prove-pythagorean-theorem-heres-what-that-means/>

- [8] Ramon Antonio, "US teens say they have new proof for 2,000-year-old mathematical theorem." From
https://www.theguardian.com/us-news/2023/mar/24/new-orleans-pythagoras-theorem-trigonometry-prove?CMP=oth_b-aplnews_d-1
- [9] MathTrain, "How High Schoolers Proved Pythagoras Using Just Trig! (and some other stuff)." From
<https://www.youtube.com/watch?v=nQD6lDwFmCc>
- [10] polymathematic, "Pythagoras Would Be Proud: High School Students' New Proof of the Pythagorean Theorem." From
<https://www.youtube.com/watch?v=p6j2nZKwf20>

Article contributed by Shailesh Shirali

Patterns in Tangent Identities

Article contributed by K Sasikumar

We are all familiar with the addition formula for the tangent function:

$$\tan(x + y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}. \quad (63)$$

By using this formula repeatedly we may derive formulas for $\tan 2x$, $\tan 3x$, $\tan 4x$, $\tan 5x$, ... in terms of $\tan x$.

After computing and closely studying these formulas, unexpected patterns get revealed. This article is about these patterns and how to explain them.

Prerequisites to understand this article

The reader needs to be familiar with the following: the principle of mathematical induction, the tangent addition formula (shown above), and the binomial identity

$$\binom{m+n}{k} = \binom{m}{0}\binom{n}{k} + \binom{m}{1}\binom{n}{k-1} + \binom{m}{2}\binom{n}{k-2} + \cdots + \binom{m}{k}\binom{n}{0}.$$

List of multiple angle tangent identities

By using the addition formula (63) repeatedly we obtain the following identities (here $t = \tan x$; we use this short-form for convenience):

$$\tan 2x = \frac{2t}{1 - t^2}$$

$$\tan 3x = \frac{3t - t^3}{1 - 3t^2}$$

$$\tan 4x = \frac{4t - 4t^3}{1 - 6t^2 + t^4}$$

$$\tan 5x = \frac{5t - 10t^3 + t^5}{1 - 10t^2 + 5t^4}$$

$$\tan 6x = \frac{6t - 20t^3 + 6t^5}{1 - 15t^2 + 15t^4 - t^6}$$

$$\tan 7x = \frac{7t - 35t^3 + 21t^5 - t^7}{1 - 21t^2 + 35t^4 - 7t^6}$$

$$\tan 8x = \frac{8t - 56t^3 + 56t^5 - 8t^7}{1 - 28t^2 + 70t^4 - 28t^6 + t^8}$$

$$\tan 9x = \frac{9t - 84t^3 + 126t^5 - 36t^7 + t^9}{1 - 36t^2 + 126t^4 - 84t^6 + 9t^8}$$

$$\tan 10x = \frac{10t - 120t^3 + 252t^5 - 120t^7 + 10t^9}{1 - 45t^2 + 210t^4 - 210t^6 + 45t^8 - t^{10}}$$

Note that we have listed the terms in the numerators and denominators in the reverse order to what we are 'used to.'

By listing the even and odd cases separately, we see the patterns clearly:

$$\tan 2x = \frac{2t}{1-t^2}$$

$$\tan 4x = \frac{4t - 4t^3}{1 - 6t^2 + t^4}$$

$$\tan 6x = \frac{6t - 20t^3 + 6t^5}{1 - 15t^2 + 15t^4 - t^6}$$

$$\tan 8x = \frac{8t - 56t^3 + 56t^5 - 8t^7}{1 - 28t^2 + 70t^4 - 28t^6 + t^8}$$

$$\tan 10x = \frac{10t - 120t^3 + 252t^5 - 120t^7 + 10t^9}{1 - 45t^2 + 210t^4 - 210t^6 + 45t^8 - t^{10}}$$

And:

$$\tan 3x = \frac{3t - t^3}{1 - 3t^2}$$

$$\tan 5x = \frac{5t - 10t^3 + t^5}{1 - 10t^2 + 5t^4}$$

$$\tan 7x = \frac{7t - 35t^3 + 21t^5 - t^7}{1 - 21t^2 + 35t^4 - 7t^6}$$

$$\tan 9x = \frac{9t - 84t^3 + 126t^5 - 36t^7 + t^9}{1 - 36t^2 + 126t^4 - 84t^6 + 9t^8}$$

Observations

Let us write,

$$\tan nx = \frac{A_n(t)}{B_n(t)}, \quad (64)$$

where $A_n(t)$ and $B_n(t)$ are polynomials in t with no common factor. (In case it appears that we have lost sight of x , let us remember that t is just a short form for $\tan x$.) Then for all n ,

- (a) $A_n(t)$ has terms only of *odd degree* (so it is an odd function).
- (b) $B_n(t)$ has terms only of *even degree* (so it is an even function).
- (c) The coefficient of t in $A_n(t)$ is n .
- (d) The coefficient of t^3 in $A_n(t)$ is $-\binom{n}{3}$.
- (e) The constant term in $B_n(t)$ is 1.
- (f) The coefficient of t^2 in $B_n(t)$ is $-\binom{n}{2}$.
- (g) This is perhaps the most surprising pattern of all.

Scan the coefficients in the expression $A_n(t)/B_n(t)$, going alternately from denominator to numerator. Note the absolute values of these coefficients.

The resulting numbers are just the binomial coefficients

$$\binom{n}{0}, \quad \binom{n}{1}, \quad \binom{n}{2}, \quad \dots \quad \binom{n}{n-2}, \quad \binom{n}{n-1}, \quad \binom{n}{n}$$

occurring in sequence. For example, consider $\tan 7x$:

$$\tan 7x = \frac{7t - 35t^3 + 21t^5 - t^7}{1 - 21t^2 + 35t^4 - 7t^6};$$

the absolute values of the coefficients are 1, 7, 21, 35, 45, 21, 7, 1. These are just the binomial coefficients $\binom{7}{0}, \binom{7}{1}, \binom{7}{2}, \dots, \binom{7}{6}, \binom{7}{7}$.

The reader will surely find many more patterns worth exploring.

A key recursion

Let us see where the tangent addition formula (63) can take us. We have,

$$\begin{aligned}\tan(n+1)x &= \frac{\frac{A_n(t)}{B_n(t)} + t}{1 - \frac{tA_n(t)}{B_n(t)}} \\ &= \frac{A_n(t) + tB_n(t)}{B_n(t) - tA_n(t)}.\end{aligned}$$

Now $\tan(n+1)x = \frac{A_{n+1}(t)}{B_{n+1}(t)}$, by definition. It therefore seems natural to conclude from the above that

$$\begin{cases} A_{n+1}(t) &= A_n(t) + tB_n(t) \\ B_{n+1}(t) &= B_n(t) - tA_n(t). \end{cases} \quad (65)$$

However, care is needed in reaching this conclusion. This is because there may be some common factors other than constants between $A_n(t) + tB_n(t)$ and $B_n(t) - tA_n(t)$.

But the statement is true; the two expressions do not have any common factors other than constants. We do not give the proof of this here as it will break the flow of the discussion; we postpone the proof to the end of the article.

We shall from this point on take the relations (65) as true and proved. Several properties now follow immediately (including many of those listed above) and may be proved using the principle of induction:

(a) *The constant term in $B_n(t)$ is 1.*

This follows from the relation $B_{n+1}(t) = B_n(t) - tA_n(t)$ along with the principle of induction. (Put $t = 0$.)

(b) *The coefficient of t in $A_n(t)$ is n .*

This follows from the relation $A_{n+1}(t) = A_n(t) + tB_n(t)$, the fact that the constant term in $B_n(t)$ is 1, and the principle of induction.

(c) *$A_n(t)$ has terms only of odd degree, and $B_n(t)$ has terms only of even degree.*

(d) *The coefficient of t^2 in $B_n(t)$ is $-\binom{n}{2}$.*

This follows from $B_{n+1}(t) = B_n(t) - tA_n(t)$, the relation

$$\binom{n+1}{2} = \binom{n}{2} + n,$$

and the principle of induction.

(e) *The coefficient of t^3 in $A_n(t)$ is $-\binom{n}{3}$.*

This follows from $A_{n+1}(t) = A_n(t) + tB_n(t)$, the relation

$$\binom{n+1}{3} = \binom{n}{3} + \binom{n}{2},$$

and the principle of induction.

By continuing this thread of reasoning a little further it is possible to get a complete formula for all the coefficients in the numerator and denominator. We leave the exploration to the reader.

Appendix

How do we justify the following claim?

$$\begin{cases} A_{n+1}(t) &= A_n(t) + tB_n(t) \\ B_{n+1}(t) &= B_n(t) - tA_n(t). \end{cases} \quad (66)$$

Here is a sketch of the proof. Suppose that there is a non-constant common factor $p(t)$ between $A_n(t) + tB_n(t)$ and $B_n(t) - tA_n(t)$. Then this common factor also divides

$$t \left(A_n(t) + tB_n(t) \right) + B_n(t) - tA_n(t) = (1 + t^2)B_n(t),$$

as well as

$$A_n(t) + tB_n(t) - t \left(B_n(t) - tA_n(t) \right) = (1 + t^2)A_n(t).$$

Therefore, $p(t)$ is a common factor of $(1 + t^2)A_n(t)$ and $(1 + t^2)B_n(t)$.

But by supposition, $A_n(t)$ and $B_n(t)$ have no non-constant common factors.

Hence $p(t)$ is a divisor of $1 + t^2$. But $1 + t^2$ cannot be factorized further without making use of complex numbers. It follows that either $p(t) = 1$ or $p(t) = 1 + t^2$.

Claim: $p(t)$ is not a factor of either $A_n(t)$ or $B_n(t)$.

To prove this we consider the roots of $A_n(t)$ and $B_n(t)$. If we can show that

$$\text{Number of real roots of } A_n(t) = \deg(A_n(t)),$$

$$\text{Number of real roots of } B_n(t) = \deg(B_n(t)),$$

then it would immediately follow that $A_n(t)$ and $B_n(t)$ cannot have non-real roots, and this would mean that $1 + t^2$ cannot be a factor of either $A_n(t)$ or $B_n(t)$.

Counting the number of real roots does not present any difficulty because these roots correspond to the angles x for which $\tan nx = 0$ (in the case of A_n) or $\cot nx = 0$ (in the case of B_n).

We leave further details to the reader. ■

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Two Problems About Primes

In this article we pose and solve two problems about prime numbers. Both the problems are very much in the ‘Olympiad tradition.’

The first problem gives the impression of being difficult but yields easily to congruence arguments (thereby showing the value of “thinking mod p ” for some prime p). One can never predict the benefits of such thinking!

The second problem highlights the value of having a good understanding of quadratic equations; e.g., knowing how to “complete the square” and knowing how to write an expression as a “difference of two squares.”

Please try your hand at the problems before reading the solutions.

Problem 1

Find all prime numbers p for which $p^2 + 23$ has exactly 10 divisors.

Problem 2

Find all prime numbers p for which $p^2 - p + 1$ is a perfect cube.

Solution to Problem 1

We are asked to find all prime numbers p for which $p^2 + 23$ has exactly 10 divisors. Following the solution given by Alon Amit [1] to a similar problem, let us examine the situation for small values of p .

p	$p^2 + 23$	# divisors of $p^2 + 23$
2	27	4
3	32	6
5	48	10
7	72	12
11	144	15
13	192	14
17	312	16
19	384	16
23	552	16
29	864	24
31	984	16
37	1392	20
41	1704	16
43	1872	30

The evidence seems to suggest that the number of divisors of $p^2 + 23$ for p prime, $p \geq 7$, is never less than 12, which (if true) would imply that there are no solutions $p \geq 7$; indeed, that the only solution is $p = 5$.

We shall show that this conclusion is true. We draw upon the known fact that if $p, q, r, \dots$ are prime numbers and $a, b, c, \dots$ are positive integers, **the number of divisors of $p^a \times q^b \times r^c \times \dots$ is $(a+1)(b+1)(c+1)\dots$** . This is a basic result in number theory; you should try proving it.

Let p be a prime number, $p \geq 5$. It has been specified that $p^2 + 23$ must have 10 divisors. Since $10 = 2 \times 5$, it follows that $p^2 + 23$ must be of the form $q \times r^4$ where q, r are prime numbers. (This is the only possibility.)

Next, observe that $23 \equiv -1 \pmod{3}$ and $23 \equiv -1 \pmod{4}$. Also, since p is prime and $p \geq 5$, it follows that p is odd and not a multiple of 3. Hence,

$$p^2 \equiv 1 \pmod{8} \quad \text{and} \quad p^2 \equiv 1 \pmod{3}, \quad (67)$$

i.e., $p^2 - 1$ is divisible by both 8 and 3 therefore by 24.

It follows that $p^2 + 23$ is a multiple of 24.

Since $24 = 2^3 \times 3$, and we know that $p^2 + 23$ must be of the form $q \times r^4$ where q, r are prime numbers, it must be that $q = 3$ and $r = 2$, and $p^2 + 23$ can only be the number $3 \times 2^4 = 48$.

Since $p^2 + 23 = 48$ when $p = 5$, it follows that there is just one solution to the problem: $p = 5$. ■

Solution to Problem 2

We are asked to find all prime numbers p for which $p^2 - p + 1$ is a perfect cube. We shall follow the elegant solution given by Brian Sittinger [2].

Before plunging in we remind the reader of a very simple-looking and but impactful observation in number theory. This is the statement that between n^2 and $(n + 1)^2$ there do not lie any perfect squares. The statement is true, as n^2 and $(n + 1)^2$ are consecutive squares. Though it appears trivial, it has significant consequences. For example, let us use it to prove that $n^2 + n + 1$ is not a perfect square for any positive integer n . The proof may be captured in a single line as follows:

If n is a positive integer, then $n^2 < n^2 + n + 1 < (n + 1)^2$.

The two inequalities above are easily checked. The conclusion follows.

We shall put this simple looking observation to use in solving the given problem.

Suppose that $p^2 - p + 1 = n^3$ for some prime number p and some integer n . It is easy to verify that the property does not hold for $p = 2, 3, 5$; hence we must have $p > 5$ and $n > 2$.

We now have,

$$\begin{aligned} p^2 - p &= n^3 - 1, \\ \therefore p(p - 1) &= (n - 1)(n^2 + n + 1). \end{aligned} \tag{68}$$

We now make use of an important property of prime numbers that is somehow not too well-known at the school level:

If a prime number p is a divisor of the product ab of two positive integers a, b , then p is a divisor of at least one of a, b .

In short:

The product of two non-multiples of a prime number p cannot be a multiple of p .

This property is not true of composite numbers. For example, 6 is the divisor of 9×10 , but 6 is not a divisor of either 9 or 10.

The above property *characterizes* prime numbers; it can be used to *define* prime numbers. But we will not go into that for now.

We invoke this property here to conclude from (68) that p must divide either $n - 1$ or $n^2 + n + 1$ or both.

We shall show that it cannot be the former.

Suppose that p is a divisor of $n - 1$; then we must have $p \leq n - 1$. Therefore, from (68) we get,

$$p - 1 \geq n^2 + n + 1.$$

So we have:

$$n^2 + n + 1 \leq p - 1 < p \leq n - 1,$$

implying that

$$n^2 + n + 1 < n - 1, \quad \text{i.e.,} \quad n^2 < -2.$$

This relation can never hold. Hence p cannot be a divisor of $n - 1$. It follows that p is a divisor of $n^2 + n + 1$. Suppose that $n^2 + n + 1 = kp$ for some positive integer k . Then we must have the following two relations,

$$n^2 + n + 1 = kp \quad \text{and} \quad k(n - 1) = p - 1. \quad (69)$$

The second relation tells us that

$$p = kn - k + 1.$$

Substituting for p in the first equation we get,

$$n^2 + n + 1 = k(kn - k + 1).$$

Hence

$$n^2 - (k^2 - 1)n + (k^2 - k + 1) = 0. \quad (70)$$

Treating this as a quadratic equation in n , we note that it is satisfied by some integer value of n and therefore has integer roots. This implies that the discriminant of the equation is a perfect square (i.e., it has an integer square root).

The discriminant is equal to

$$(k^2 - 1)^2 - 4(k^2 - k + 1) = k^4 - 6k^2 + 4k - 3. \quad (71)$$

Therefore, $k^4 - 6k^2 + 4k - 3$ must be a perfect square. So, let us first find all the integer values of k for which $k^4 - 6k^2 + 4k - 3$ is a perfect square.

The values taken by $k^4 - 6k^2 + 4k - 3$ for $k = 1, 2, 3, 4, \dots$ are given below:

k	1	2	3	4	5	6	7	8	9
$k^4 - 6k^2 + 4k - 3$	-4	-3	36	173	492	1101	2132	3741	6108

We quickly spot the perfect square 36 which is one of the values taken by the given expression (for $k = 3$). Is it possible that this is the only such square?

To explore this, let us try to 'box' $k^4 - 6k^2 + 4k - 3$ between two perfect squares (following the comment made at the start of this write-up).

Observe that

$$(k^2 - 3)^2 = k^4 - 6k^2 + 9,$$

and so,

$$(k^4 - 6k^2 + 4k - 3) - (k^2 - 3)^2 = 4k - 12 = 4(k - 3).$$

This implies that if $k > 3$ then

$$(k^2 - 3)^2 < k^4 - 6k^2 + 4k - 3. \quad (72)$$

Also,

$$(k^2 - 2)^2 = k^4 - 4k^2 + 4,$$

and so,

$$(k^4 - 6k^2 + 4k - 3) - (k^2 - 2)^2 = -2k^2 + 4k - 7.$$

It is not difficult to see that

$$-2k^2 + 4k - 7 < 0 \quad \text{for all } k.$$

For,

$$\begin{aligned} -2k^2 + 4k - 7 &= -2(k^2 - 2k) - 7 \\ &= -2(k^2 - 2k + 1) - 5 \\ &= -2(k - 1)^2 - 5 \leq -5 \quad \text{for all } k. \end{aligned}$$

This implies that

$$k^4 - 6k^2 + 4k - 3 < (k^2 - 2)^2 \quad \text{for all } k. \quad (73)$$

It follows that if $k > 3$,

$$(k^2 - 3)^2 < k^4 - 6k^2 + 4k - 3 < (k^2 - 2)^2. \quad (74)$$

We conclude that $k^4 - 6k^2 + 4k - 3$ cannot be a perfect square for any $k > 3$.

Therefore the only values of k which we need to check are $k = 1, 2, 3$.

The values taken by $k^4 - 6k^2 + 4k - 3$ for $k = 1, 2, 3$ are (respectively) $-4, -3, 36$. We see that the only square value taken is 36, when $k = 3$.

For $k = 3$, the quadratic equation (70) reduces to

$$n^2 - 8n + 7 = 0,$$

and we obtain the solutions $n = 1, 7$. The solution $n = 1$ is not meaningful here, as you can verify (it yields $p = 1$). And we can also verify that $n = 7$ does correspond to a genuine solution. For, from $n = 7$ and $k = 3$ we get

$$n^2 + n + 1 = 49 + 7 + 1 = 57, \quad \frac{n^2 + n + 1}{k} = \frac{57}{3} = 19,$$

and we have:

$$19^2 - 19 + 1 = 343 = 7^3.$$

Further reading

- [1] Alon Amit, "What's the sum of the numbers p in the set of primes such that $p^2 + 11$ has exactly six different positive divisors?" from <https://tinyurl.com/3fmjaavx>
- [2] Brian Sittinger, "How do I find all primes p such that $p^2 - p + 1$ is a perfect cube?" from <https://tinyurl.com/5n6wvznd>

Goldbach Reversed!

Goldbach's Conjecture, And How We Were Led To Modify It

We would guess that most readers have heard of Goldbach's Conjecture, which originated in a letter written by Christian Goldbach to Leonhard Euler in 1742. In his reply Euler modified the conjecture, and it is in this modified form that it is generally known today (though by name it is attributed solely to Goldbach). We state it in the following form:

***Goldbach Conjecture.** Every even number greater than 4 may be written as the sum of two odd primes.*

For example, we have:

$$6 = 3 + 3,$$

$$8 = 3 + 5,$$

$$10 = 3 + 7,$$

$$12 = 5 + 7,$$

$$60 = 7 + 53,$$

$$100 = 11 + 89,$$

and so on.

Note that since $4 = 2 + 2$ too can be written as the sum of two primes, we could instead say “every even number greater than 2” and dispense with the word “odd”. But we shall let it stay, as it has to do with the theme of this article.

Goldbach’s Conjecture has a deceptively simple look about it. Generations of mathematicians have tried their hand at proving it, only to taste defeat. This article is written to bring some cheer to them! For this, we first recall an important principle in problem solving pedagogy first stated by George Polyá:

Polyá Principle. *If a problem $\mathcal{P}$ seems too tough to solve, then tackle some suitable problem $\mathcal{P}'$ that can be obtained as a special case of $\mathcal{P}$ — the tougher the better, and which is in your power to solve.*

It may happen that even this is too ambitious a plan! So, with due apologies to Polyá, we shall replace this with a modified principle which now guarantees success:

Modified Polyá Principle. *If a problem $\mathcal{P}$ seems too tough to solve, then tackle some suitable problem $\mathcal{P}''$ that is related in some way to $\mathcal{P}$, at least superficially, and which is in your power to solve.*

Using this hugely empowering principle to cheer us on, we produce a modified version of the Goldbach conjecture:

Modified Goldbach Conjecture. *Every even number greater than 4 may be written as the sum of two odd, composite numbers.*

But this does not look too promising; why, we cannot even write 4 as a sum of two odd, composite numbers! Nor can we so write 6; nor 8; nor 10. Indeed, the first even number that can be so written is $18 = 9 + 9$ (we allow a number to be used twice in the sum), and the next one is $24 = 9 + 15$.

So our Modified Goldbach Conjecture seems to have met a speedy end. But let us not throw it out so quickly! Persevering a little longer, we find to our surprise that the even numbers that *cannot* be written as sums of two odd, composite numbers appear to be limited in number! Indeed, the numbers appear to be precisely the following:

2, 4, 6, 8, 10, 12, 14, 16, 20, 22, 26, 28, 32, 38.

Please check for yourself that none of these numbers can be written as the sum of two odd, composite numbers.

What is not immediately obvious is that the list displayed above cannot be continued; it comes to an abrupt end! Indeed, we find that the following is a true statement: *Every even number from 40 onwards can be written as a sum of two odd, composite numbers.* So we have been able to extract a little nugget from this ramble!

We now state and prove the main result of this article.

Theorem. *Every even number greater than or equal to 40 can be written as a sum of two odd, composite numbers. The largest even number that cannot be so written is 38.*

Proof of the theorem

The proof is not hard to find. We use the following:

Lemma. *Let d be a positive integer not divisible by 3. Let a be a positive integer larger than 3. Then it is not possible for all three of the numbers a , $a + d$, $a + 2d$ to be prime.*

In short, three distinct integers in arithmetic progression (AP for short) with a common difference not divisible by 3 cannot all be prime numbers.

This is so because the conditions force one of $a, a + d, a + 2d$ to be a multiple of 3. As this number exceeds 3, it must be composite.

To make use of this fact, we use the additional fact that 15, 25, 35 are odd, composite integers in AP, the common difference being 10.

Let $n \geq 40$ be an even number. We consider the three odd numbers

$$n - 35, \quad n - 25, \quad n - 15$$

which are in AP; the common difference is 10, which is not divisible by 3. The least of the numbers exceeds 3, so they cannot all be prime; therefore, one of them is composite; and, of course, it is odd. Tracing back, we get an expression for n as a sum of two odd, composite numbers.

This proof fails for $n = 38$, precisely because $38 - 35$ does not exceed 3. It is easy to check, empirically, that 38 cannot be written as a sum of two odd, composite numbers. 

Closing remarks

On a more serious closing note, we remark that Polyá-style ‘relaxations’ of the Goldbach conjecture *have* been proved, typically involving very deep ideas and a great deal of effort; for example:

- (A) *Every even integer is the sum of at most six primes.* (Proved by Ramaré in 1995.)
- (B) *Every sufficiently large even integer is the sum of a prime and a number with no more than two prime factors.* (Proved by Chen in 1966.)
- (C) *Every sufficiently large odd integer is the sum of three odd primes.* (Proved by Vinogradov in 1937.)

Note the “relaxations” in these statements when compared with the original formulation of the conjecture:

- In (A) we permit the use of up to *six* primes.
- In (B), we allow one of the summands to be a number with up to *two* prime factors, and we make the claim only for *sufficiently large numbers*.
- As regards (C), note that Vinogradov’s result is implied by the Goldbach result (if it is true), but the reverse cannot be claimed; so (C) is indeed weaker than Goldbach’s statement.

At a much simpler level is the following:

- (D) Given any integer $n > 1$, and any even positive integer k , primes p and q exist such that $p + q \equiv k \pmod{n}$

Statement (D) may be called a “modular” form of Goldbach’s conjecture; but unlike that conjecture, it can be proved!

Article contributed by the editors

Call for Articles

We invite mathematics teachers and practitioners to contribute short articles to The Compass.

Articles could deal with mathematical content or the teaching of mathematics (classes 6 to 12). They could feature real-life classroom experiences with projects and explorations.

They could be about the use of audiovisual material or social media in the mathematics classroom.

They could feature reflections on the teaching of mathematics, or on the nature of mathematics.

Articles about the use of software are particularly welcome, as are articles dealing with non-routine problems and classroom experiences with problem-solving.

In short, anything at all to do with school mathematics!